

EBOOK PANDUAN LENGKAP PENGEMBANGAN SISTEM E-PRESENSI GTK MADRASAH BERBASIS NEXT.JS 16 & PRISMA

Panduan Super Detail dari Nol: Arsitektur Multi-Tenant, Geofence GPS, Swafoto Watermark, PWA Native, Dashboard Monitoring Admin, Rekap EMIS 4.0 & Panduan Tambah Modul Baru
Disertai Kode Sumber Utuh (Full Complete Code) dan Dokumentasi Baris per Baris

Spesifikasi & Cakupan Blueprint Sistem

- **Framework & Runtime:** Next.js 16 (App Router, Server Actions, RSC) + Node.js 20+ LTS
- **Database & ORM:** PostgreSQL (Supabase Pooler) + Prisma ORM 6.4 + Indeks Komposit
- **Desain & Styling:** Tailwind CSS v4 + Plus Jakarta Sans Typography + Lucide Icons
- **Autentikasi & Keamanan:** NextAuth.js v5 Beta (Credentials Multi-ID: NIP, NIK, NUPTK, PegID)
- **Modul Mobile GTK:** PWA Standalone (Service Worker, Geofencing Haversine, Kamera Live)
- **Format Standar Dinas:** Laporan Rekapitulasi Presensi EMIS GTK 4.0 Kemenag (.xlsx & .pdf)
- **Format Dokumen Ini:** Kertas F4 / Folio Standar Resmi Indonesia (215 mm x 330 mm)

DOKUMENTASI RESMI REKAYASA PERANGKAT LUNAK E-PRESENSI V2

Disusun secara lengkap menyertakan seluruh kode implementasi produksi, arsitektur skema, algoritma geofencing, penempelan watermark canvas, hingga panduan pembuatan modul baru dari nol.
Edisi Lengkap F4 (Folio) | Kode Sumber Utuh | Tahun 2026

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STANDARISASI KODE SUMBER UTUH (FULL SOURCE CODE)

Seluruh blok kode dalam ebook ini diambil langsung dari codebase proyek nyata (Next.js 16 + Prisma 6 + PostgreSQL). Setiap file kode disajikan secara utuh tanpa pemotongan (...) agar pembaca dapat langsung menyalin (copy-paste) atau menelaah arsitektur logika secara menyeluruh.

BAB 1

Pengenalan Sistem, Latar Belakang & Arsitektur Solusi

Memahami landasan arsitektur sistem informasi kehadiran terintegrasi, pemisahan hak akses multi-tenant, dan diagram alur interaksi antarmuka web serta mobile GTK.

1.1 Latar Belakang & Urgensi Digitalisasi Presensi GTK

Sistem E-Presensi GTK Madrasah V2 merupakan solusi perangkat lunak terintegrasi yang dirancang untuk memodernisasi tata kelola absensi guru dan tenaga kependidikan di lingkungan Kementerian Agama RI. Metode konvensional seperti absensi manual berbasis tanda tangan kertas memiliki kelemahan mendasar: tingginya potensi kecurangan waktu (titip absen), lambatnya perekapan bulanan oleh bagian Tata Usaha, dan hilangnya rekaman fisik.

Sistem ini menerapkan validasi ganda berbasis geospasial (Geofencing GPS) dan penangkapan swafoto live camera dengan watermarking otomatis (nama, NIP, koordinat bujur-lintang, jarak dari gerbang madrasah, dan penanda waktu WIB). Seluruh data disinkronkan secara real-time ke pusat kendali administrator madrasah.

1.2 Hirarki Peran Pengguna (Role-Based Access Control)

Peran (Role)	Cakupan Akses (Scope)	Hak Istimewa & Fitur Utama
SUPERADMIN	Global (Lintas Seluruh Madrasah)	Mengelola data master madrasah, pengaturan global sistem, audit log aktivitas kepegawaian lintas satuan kerja.
ADMIN_MADRASAH	Satuan Kerja Madrasah Tertentu	Monitoring absensi real-time, verifikasi pengajuan izin/sakit, input presensi massal, cetak laporan resmi EMIS 4.0.
TEACHER / GTK	Pribadi Guru (Self-Service)	Presensi masuk/pulang dalam radius geofence, pengisian jurnal KBM harian, pengajuan cuti, riwayat absensi pribadi.

1.3 Diagram Interaksi Arsitektur (Architecture Flow)

Aplikasi menggunakan paradigma Next.js 16 App Router dengan React Server Components (RSC) dan Server Actions. Server Actions menggantikan REST API tradisional, memungkinkan pemanggilan fungsi backend secara langsung dan terproteksi tipe data (Type-Safe) dari komponen antarmuka klien.

BAB 2

Persiapan Environment & Inisialisasi Proyek dari Nol

Langkah-langkah mendasar instalasi perangkat lunak, inisialisasi Next.js 16, konfigurasi Tailwind CSS v4, dan perancangan struktur folder berbasis Feature-First Architecture.

2.1 Prasyarat Sistem & Konfigurasi Terminal

Pastikan Node.js v20+ atau v22+ telah terpasang dengan baik pada komputer pengembangan:

```
KODE LENGKAP: terminal / powershell
1 # Verifikasi versi runtime
2 node -v # Output minimal: v20.10.0 atau lebih baru
3 npm -v # Output minimal: 10.x.x
4 git --version
```

2.2 Inisialisasi Next.js 16 App Router

```
KODE LENGKAP: terminal / powershell
1 npx create-next-app@latest e-presensi-v2 --typescript --eslint --tailwind --app --src-...
2 cd e-presensi-v2
```

2.3 Instalasi Dependensi Pustaka Lengkap

```
KODE LENGKAP: package.json installation
1 # 1. Database & ORM
2 npm install @prisma/client
3 npm install -D prisma tsx
4
5 # 2. Autentikasi Kepegawaian & Kriptografi
6 npm install next-auth@beta bcryptjs zod
7 npm install -D @types/bcryptjs
8
9 # 3. Komponen Antarmuka & Ikonografi
10 npm install lucide-react recharts sweetalert2 clsx tailwind-merge
11
12 # 4. Ekspor Laporan Excel & PDF Resmi Dinas
13 npm install xlsx jspdf jspdf-autotable
```

2.4 Konfigurasi package.json Lengkap

```
KODE LENGKAP: package.json
1 {
2   "name": "web",
3   "version": "0.1.0",
4   "private": true,
5   "scripts": {
6     "dev": "next dev",
7     "build": "prisma generate && next build",
8     "start": "next start",
9     "lint": "eslint",
10    "generate:ebook": "node scripts/generate-ebook-f4.js",
11    "postinstall": "prisma generate"
12  },
13  "dependencies": {
14    "@base-ui/react": "^1.8.0",
15    "@hookform/resolvers": "^5.9.1",
16    "@prisma/client": "^6.4.1",
17    "@tanstack/react-query": "^5.104.0",
18    "@tanstack/react-table": "^9.2.4",
19    "axios": "^1.20.0",
20    "bcryptjs": "^3.0.3",
21    "class-variance-authority": "^0.7.1",
```

22	"clsx": "^2.1.1",
23	"cn": "^0.4.0",
24	"date-fns": "^4.4.0",
25	"html2canvas": "^1.4.1",
26	"jspdf": "^4.2.1",
27	"jspdf-autotable": "^5.0.8",
28	"lucide-react": "^1.48.0",
29	"next": "16.3.6",
30	"next-auth": "^5.0.0-beta.32",
31	"react": "19.2.8",
32	"react-dom": "19.2.8",
33	"react-hook-form": "^7.89.0",
34	"recharts": "^3.10.1",
35	"shadcn": "^4.21.0",
36	"sweetalert2": "^11.26.25",
37	"tailwind-merge": "^3.7.0",
38	"tw-animate-css": "^1.4.0",
39	"xlsx": "^0.18.5",
40	"zod": "^3.25.76"
41	},
42	"devDependencies": {
43	"@tailwindcss/postcss": "^4",
44	"@types/bcryptjs": "^2.4.6",
45	"@types/node": "^20",
46	"@types/react": "^19",
47	"@types/react-dom": "^19",
48	"eslint": "^9",
49	"eslint-config-next": "16.3.6",
50	"prisma": "^6.4.1",
51	"tailwindcss": "^4",
52	"tsx": "^4.23.15",
53	"typescript": "^5"
54	},
55	"prisma": {
56	"seed": "tsx prisma/seed.ts"
57	}
58	}

BAB 3

Perancangan Basis Data PostgreSQL dengan Prisma ORM

Mendesain skema basis data relasional multi-tenant yang tangguh, menerapkan constraint integritas unik, tipe data PostgreSQL, serta konfigurasi indeks performa tinggi.

3.1 Kode Sumber Utuh Skema Basis Data (schema.prisma)

Berikut adalah isi lengkap berkas `prisma/schema.prisma` yang memuat seluruh model data, relasi multi-tenant, serta indeks komposit untuk optimasi kueri tingkat tinggi:

```
KODE | ENCAP: prisma/schema.prisma
1  generator client {
2    provider = "prisma-client-js"
3  }
4
5  datasource db {
6    provider = "postgresql"
7    url       = env("DATABASE_URL")
8    directUrl = env("DIRECT_URL")
9  }
10
11 // =====
12 // ENUMS
13 // =====
14
15 enum Role {
16   SUPERADMIN
17   ADMIN_MADRASAH
18   TEACHER
19 }
20
21 enum AttendanceStatus {
22   PRESENT
23   LATE
24   PERMIT
25   SICK
26   ABSENT
27 }
28
29 // =====
30 // MODELS
31 // =====
32
33 model Madrasah {
34   id          String    @id @default(uuid())
35   nsm         String    @unique // Nomor Statistik Madrasah
36   npsn        String?   @unique // Nomor Pokok Sekolah Nasional
37   name        String
38   address     String?   @db.Text
39   phone       String?
40   email       String?
41   isActive    Boolean   @default(true)
42   createdAt   DateTime  @default(now()) @map("created_at")
43   updatedAt   DateTime  @updatedAt @map("updated_at")
44
45   // Relations
46   settings    MadrasahSetting?
47   users       User[]
48   attendanceLogs AttendanceLog[]
49   holidays    Holiday[]
50   positions    Position[]
51   teachingJournals TeachingJournal[]
52
53   @@map("madrasahs")
54 }
55
56 model MadrasahSetting {
57   id          String    @id @default(uuid())
```

58	madrasahId	String	@unique @map("madrasah_id")
59			
60	// Geofencing Coordinates		
61	latitude	Float	
62	longitude	Float	
63	radiusMeters	Float	@default(50.0) @map("radius_meters") // Default radius 50m
64			
65	// Attendance Work Hours (HH:mm format)		
66	workStartTime	String	@default("07:00") @map("work_start_time")
67	lateThreshold	String	@default("07:15") @map("late_threshold")
68	workEndTime	String	@default("14:00") @map("work_end_time")
69			
70	// Work Days & Daily Specific Hours (JSON String for Monday to Sunday)		
71	workDays	String	@default("1,2,3,4,5") @map("work_days")
72	dailySchedules	String?	@db.Text @map("daily_schedules")
73	requireSelfie	Boolean	@default(true) @map("require_selfie")
74	allowBackdatedAttendance	Boolean	@default(false) @map("allow_backdated_attendance")...
75			
76	createdAt	DateTime	@default(now()) @map("created_at")
77	updatedAt	DateTime	@updatedAt @map("updated_at")
78			
79	// Relations		
80	madrasah	Madrasah	@relation(fields: [madrasahId], references: [id], onDelete:...
81			
82	@@map("madrasah_settings")		
83	}		
84			
85	model Holiday {		
86	id	String	@id @default(uuid())
87	madrasahId	String	@map("madrasah_id")
88	name	String	
89	date	DateTime	@db.Date
90	endDate	DateTime?	@db.Date @map("end_date") // Optional for multi-day long vacat...
91	isNational	Boolean	@default(false) @map("is_national")
92	description	String?	@db.Text
93	createdAt	DateTime	@default(now()) @map("created_at")
94			
95	// Relations		
96	madrasah	Madrasah	@relation(fields: [madrasahId], references: [id], onDelete: Ca...
97			
98	@@index([madrasahId])		
99	@@index([madrasahId, date])		
100	@@index([madrasahId, endDate])		
101	@@map("holidays")		
102	}		
103			
104	model Position {		
105	id	String	@id @default(uuid())
106	madrasahId	String	@map("madrasah_id")
107	name	String	// e.g. "Kepala Madrasah", "Wakil Kepala", "Guru Kelas", "Guru...
108	code	String?	// e.g. "KAMAD", "WAKAMAD", "GURU", "TU"
109	description	String?	@db.Text
110	isHeadmaster	Boolean	@default(false) @map("is_headmaster") // Penanda khusus Kepala...
111	order	Int	@default(0)
112	createdAt	DateTime	@default(now()) @map("created_at")
113	updatedAt	DateTime	@updatedAt @map("updated_at")
114			
115	// Relations		
116	madrasah	Madrasah	@relation(fields: [madrasahId], references: [id], onDelete: Ca...
117	users	User[]	
118			
119	@@unique([madrasahId, name])		
120	@@index([madrasahId])		
121	@@index([madrasahId, order])		
122	@@map("positions")		
123	}		
124			
125	model User {		
126	id	String	@id @default(uuid())
127	madrasahId	String?	@map("madrasah_id") // Nullable ONLY for SUPERADMIN
128	positionId	String?	@map("position_id")
129	name	String	
130	gelarDepan	String?	@map("gelar_depan")
131	gelarBelakang	String?	@map("gelar_belakang")
132	email	String	@unique
133	nip	String?	// Nomor Induk Pegawai

134	nik	String?	// Nomor Induk Kependudukan (16 digit)
135	pegId	String?	@map("peg_id") // ID Pegawai Simpatika / EMIS Kemenag
136	nuptk	String?	// Nomor Unik Pendidik & Tenaga Kependidikan (16 digit)
137	tempatLahir	String?	@map("tempat_lahir")
138	tanggalLahir	DateTime?	@db.Date @map("tanggal_lahir")
139	gender	String?	// L (Laki-laki) / P (Perempuan)
140	statusKepegawaian	String?	@map("status_kepegawaian") // PNS / PPPK / Non-PNS / GTY...
141	jenisGtk	String?	@map("jenis_gtk") // Guru Mapel / Guru Kelas / Tenaga Ad...
142	passwordHash	String	@map("password_hash")
143	role	Role	@default(TEACHER)
144	phone	String?	
145	avatarUrl	String?	@map("avatar_url")
146	isActive	Boolean	@default(true) @map("is_active")
147	createdAt	DateTime	@default(now()) @map("created_at")
148	updatedAt	DateTime	@updatedAt @map("updated_at")
149			
150	// Relations		
151	madrasah	Madrasah?	@relation(fields: [madrasahId], references: [id], on...
152	position	Position?	@relation(fields: [positionId], references: [id], on...
153	attendanceLogs	AttendanceLog[]	
154	teachingJournals	TeachingJournal[]	
155			
156	@@index([madrasahId])		
157	@@index([positionId])		
158	@@index([role])		
159	@@index([pegId])		
160	@@index([nuptk])		
161	@@index([nip])		
162	@@index([nik])		
163	@@index([name])		
164	@@index([madrasahId, role, isActive])		
165	@@map("users")		
166	}		
167			
168	model AttendanceLog {		
169	id	String	@id @default(uuid())
170	madrasahId	String	@map("madrasah_id") // Multi-tenant strict isolat...
171	userId	String	@map("user_id")
172	date	DateTime	@db.Date // YYYY-MM-DD for single check per day c...
173			
174	// Clock-in Details		
175	checkInTime	DateTime?	@map("check_in_time")
176	checkInLat	Float?	@map("check_in_lat")
177	checkInLng	Float?	@map("check_in_lng")
178	checkInDistance	Float?	@map("check_in_distance") // Distance in meters f...
179	checkInPhotoUrl	String?	@map("check_in_photo_url")
180			
181	// Clock-out Details		
182	checkOutTime	DateTime?	@map("check_out_time")
183	checkOutLat	Float?	@map("check_out_lat")
184	checkOutLng	Float?	@map("check_out_lng")
185	checkOutDistance	Float?	@map("check_out_distance")
186	checkOutPhotoUrl	String?	@map("check_out_photo_url")
187			
188	status	AttendanceStatus	@default(PRESENT)
189	notes	String?	@db.Text
190	createdAt	DateTime	@default(now()) @map("created_at")
191	updatedAt	DateTime	@updatedAt @map("updated_at")
192			
193	// Relations		
194	madrasah	Madrasah	@relation(fields: [madrasahId], references: [id],...
195	user	User	@relation(fields: [userId], references: [id], onD...
196			
197	@@unique([userId, date]) // 1 user has at most 1 attendance record per calendar date		
198	@@index([madrasahId])		
199	@@index([madrasahId, date])		
200	@@index([userId])		
201	@@index([madrasahId, status])		
202	@@index([madrasahId, date, status])		
203	@@map("attendance_logs")		
204	}		
205			
206	model TeachingJournal {		
207	id	String	@id @default(uuid())
208	madrasahId	String	@map("madrasah_id")
209	userId	String	@map("user_id")

210	date	DateTime	@db.Date // YYYY-MM-DD
211	className	String	@map("class_name") // e.g. "X MIPA 1", "VII A", "Kelas 1", ...
212	subjectName	String	@map("subject_name") // e.g. "Matematika", "Fiqih", "Bahasa..."
213	sessionHours	String	@map("session_hours") // e.g. "Jam 1-2 (07:30 - 09:00)"
214	topicTitle	String	@map("topic_title") // Materi Pokok / Bahasan
215	activities	String	@db.Text // Uraian Aktivitas KBM
216	studentPresence	String?	@map("student_presence") // e.g. "Hadir: 32, Izin: 1, Alpha:...
217	notes	String?	@db.Text // Catatan hambatan / tindak lanjut
218	photoUrl	String?	@map("photo_url") // Foto dokumentasi KBM
219	createdAt	DateTime	@default(now()) @map("created_at")
220	updatedAt	DateTime	@updatedAt @map("updated_at")
221			
222	// Relations		
223	madrasah	Madrasah	@relation(fields: [madrasahId], references: [id], onDelete:...
224	user	User	@relation(fields: [userId], references: [id], onDelete: Cas...
225			
226	@@index([madrasahId])		
227	@@index([madrasahId, date])		
228	@@index([userId])		
229	@@index([userId, date])		
230	@@index([madrasahId, className])		
231	@@map("teaching_journals")		
232	}		

3.2 Kode Sumber Seeder Basis Data Awal (prisma/seed.ts)

File seeder ini bertugas mengisi data awal (Superadmin Kemenag, Madrasah MIN 1 Jakarta Selatan, Admin Madrasah, dan akun GTK percontohan):

```
KODE LENGKAP: prisma/seed.ts
1  import { PrismaClient } from "@prisma/client";
2  import bcrypt from "bcryptjs";
3
4  const prisma = new PrismaClient();
5
6  async function main() {
7    console.log("ðŸš€ Memulai seeding database...");
8
9    // 1. Password default terenkripsi
10   const defaultPasswordHash = await bcrypt.hash("Password123!", 10);
11
12   // 2. Buat Superadmin Kemenag
13   const superadmin = await prisma.user.upsert({
14     where: { email: "superadmin@kemenag.go.id" },
15     update: {},
16     create: {
17       name: "Super Administrator Kemenag",
18       email: "superadmin@kemenag.go.id",
19       passwordHash: defaultPasswordHash,
20       role: "SUPERADMIN",
21       nip: "198001012005011001",
22       phone: "081234567890",
23       isActive: true,
24     },
25   });
26   console.log(`Superadmin dibuat: ${superadmin.email}`);
27
28   // 3. Buat Data Madrasah Contoh: MIN 1 Jakarta Selatan
29   const madrasah = await prisma.madrasah.upsert({
30     where: { nsm: "111131740001" },
31     update: {},
32     create: {
33       name: "MIN 1 Jakarta Selatan",
34       nsm: "111131740001",
35       npsn: "60721234",
36       address: "Jl. Madrasah No. 1, Cilandak, Jakarta Selatan",
37       phone: "021-7654321",
38       email: "kontak@min1jaksel.sch.id",
39       isActive: true,
40       settings: {
41         create: {
42           latitude: -6.2615,
43           longitude: 106.8106,
```

```
44     radiusMeters: 50.0,
45     workStartTime: "07:00",
46     lateThreshold: "07:15",
47     workEndTime: "14:00",
48   },
49   },
50   },
51   include: {
52     settings: true,
53   },
54 });
55   console.log(`  Madrasah dibuat: ${madrasah.name} (Radius G
56
57 // 4. Buat Admin Madrasah (Operator)
58 const adminMadrasah = await prisma.user.upsert({
59   where: { email: "admin@min1jaksel.sch.id" },
60   update: {},
61   create: {
62     name: "Operator Madrasah MIN 1",
63     email: "admin@min1jaksel.sch.id",
64     passwordHash: defaultPasswordHash,
65     role: "ADMIN_MADRASAH",
66     madrasahId: madrasah.id,
67     nip: "198805202012011003",
68     phone: "081298765432",
69     isActive: true,
70   },
71 });
72   console.log(`  Admin Madrasah dibuat: ${adminMadrasah.ema i
73
74 // 5. Buat Guru Contoh & Rekam Kehadiran
75 const teachersData = [
76   {
77     name: "Ahmad Fauzi, S.Pd.I",
78     email: "fauzi@min1jaksel.sch.id",
79     nip: "199203152019031002",
80     phone: "081311223344",
81     status: "PRESENT" as const,
82     checkInTime: new Date(new Date().setHours(6, 52, 0, 0)),
83     checkInDistance: 18.5,
84     checkInLat: -6.26148,
85     checkInLng: 106.81062,
86     notes: "Tepat waktu di area kelas 3A",
87   },
88   {
89     name: "Siti Nurhaliza, M.Pd",
90     email: "siti@min1jaksel.sch.id",
91     nip: "198711042014022001",
92     phone: "081299887766",
93     status: "PRESENT" as const,
94     checkInTime: new Date(new Date().setHours(7, 5, 0, 0)),
95     checkInDistance: 24.1,
96     checkInLat: -6.26152,
97     checkInLng: 106.81055,
98     notes: "Hadir tepat waktu sebelum apel pagi",
99   },
100  {
101    name: "Drs. H. Muhammad Ridwan",
102    email: "ridwan@min1jaksel.sch.id",
103    nip: "197508122002121003",
104    phone: "081234123412",
105    status: "LATE" as const,
106    checkInTime: new Date(new Date().setHours(7, 24, 0, 0)),
107    checkInDistance: 32.7,
108    checkInLat: -6.26155,
109    checkInLng: 106.81045,
110    notes: "Terlambat 9 menit karena kendala lalu lintas",
111  },
112  {
113    name: "Fathimatuzzahra, S.Pd",
114    email: "zahra@min1jaksel.sch.id",
115    nip: "199501252020122004",
116    phone: "085678912345",
117    status: "PERMIT" as const,
118    checkInTime: null,
119    checkInDistance: null,
```

```
120     checkInLat: null,
121     checkInLng: null,
122     notes: "Izin dinas: Menghadiri Pelatihan Kurikulum Kemenag di Balai Diklat",
123   },
124   {
125     name: "Budi Santoso, S.Kom",
126     email: "budi@min1jaksel.sch.id",
127     nip: "199009182018011002",
128     phone: "087788990011",
129     status: "SICK" as const,
130     checkInTime: null,
131     checkInDistance: null,
132     checkInLat: null,
133     checkInLng: null,
134     notes: "Sakit (Surat Dokter No. 142/KLN/IX/2026)",
135   },
136 ];
137
138 const todayDate = new Date();
139 todayDate.setHours(0, 0, 0, 0);
140
141 for (const t of teachersData) {
142   const user = await prisma.user.upsert({
143     where: { email: t.email },
144     update: { name: t.name, nip: t.nip, phone: t.phone },
145     create: {
146       name: t.name,
147       email: t.email,
148       passwordHash: defaultPasswordHash,
149       role: "TEACHER",
150       madrasahId: madrasah.id,
151       nip: t.nip,
152       phone: t.phone,
153       isActive: true,
154     },
155   });
156
157   await prisma.attendanceLog.upsert({
158     where: {
159       userId_date: {
160         userId: user.id,
161         date: todayDate,
162       },
163     },
164     update: {
165       status: t.status,
166       checkInTime: t.checkInTime,
167       checkInDistance: t.checkInDistance,
168       checkInLat: t.checkInLat,
169       checkInLng: t.checkInLng,
170       notes: t.notes,
171     },
172     create: {
173       madrasahId: madrasah.id,
174       userId: user.id,
175       date: todayDate,
176       status: t.status,
177       checkInTime: t.checkInTime,
178       checkInDistance: t.checkInDistance,
179       checkInLat: t.checkInLat,
180       checkInLng: t.checkInLng,
181       notes: t.notes,
182     },
183   });
184 }
185
186 // 6. Buat Hari Libur Contoh
187 const holidays = [
188   {
189     name: "Hari Santri Nasional",
190     date: new Date("2026-10-22"),
191     description: "Peringatan Hari Santri Nasional Kementerian Agama RI",
192   },
193   {
194     name: "Hari Guru Nasional",
195     date: new Date("2026-11-25"),
```

```
196     description: "Apresiasi Guru & Tenaga Kependidikan Madrasah Ibtidaiyah",
197   },
198   {
199     name: "Cuti Bersama Akhir Semester Ganjil",
200     date: new Date("2026-12-24"),
201     description: "Libur Pembagian Rapor & Evaluasi Pembelajaran",
202   },
203 ];
204
205 for (const h of holidays) {
206   const existing = await prisma.holiday.findFirst({
207     where: { madrasahId: madrasah.id, name: h.name },
208   });
209   if (!existing) {
210     await prisma.holiday.create({
211       data: {
212         madrasahId: madrasah.id,
213         name: h.name,
214         date: h.date,
215         description: h.description,
216       },
217     });
218   }
219 }
220 console.log(` 3 Hari Libur Resmi Madrasah berhasil disia p
221
222 console.log(" \n0%B Seeding selesai dengan sukses! ");
223 console.log("Kredensial Default Login (Password: Password123!):");
224 console.log("1. Superadmin      : superadmin@kemenag.go.id");
225 console.log("2. Admin Madrasah : admin@min1jaksel.sch.id");
226 console.log("3. Guru (Mobile)  : fauzi@min1jaksel.sch.id / NIP: 199203152019031002")...
227 }
228
229 main()
230 .catch((e) => {
231   console.error("Gagal melakukan seeding:", e);
232   process.exit(1);
233 })
234 .finally(async () => {
235   await prisma.$disconnect();
236 });
```

BAB 4

Autentikasi Kepegawaian & Otorisasi Rute (NextAuth.js v5)

Membangun sistem otentikasi aman berbasis NextAuth v5 Beta dengan dukungan multi-identifier, enkripsi bcrypt, token session JWT, dan middleware proteksi rute ganda.

4.1 Konfigurasi Penyedia Kredensial Multi-ID (src/lib/auth.ts)

Berikut adalah kode sumber utuh `src/lib/auth.ts` yang menangani login dengan NIP, NIK, NUPTK, PegID, maupun email resmi:

```
KODE SUMBER: src/lib/auth.ts
1  import NextAuth from "next-auth";
2  import Credentials from "next-auth/providers/credentials";
3  import bcrypt from "bcryptjs";
4  import { z } from "zod";
5  import { prisma } from "@/lib/prisma";
6
7  const loginSchema = z.object({
8    email: z.string().min(3, "Masukkan email atau NIP valid"),
9    password: z.string().min(1, "Password wajib diisi"),
10  });
11
12  export const { handlers, signIn, signOut, auth } = NextAuth({
13    trustHost: true,
14    providers: [
15      Credentials({
16        name: "Credentials",
17        credentials: {
18          email: { label: "Email / NIP", type: "text" },
19          password: { label: "Password", type: "password" },
20        },
21        async authorize(credentials) {
22          const parsed = loginSchema.safeParse(credentials);
23          if (!parsed.success) {
24            return null;
25          }
26
27          const identifier = parsed.data.email.trim();
28          const user = await prisma.user.findFirst({
29            where: {
30              OR: [
31                { email: { equals: identifier, mode: "insensitive" } },
32                { nip: identifier },
33                { nuptk: identifier },
34                { pegId: identifier },
35                { nik: identifier },
36              ],
37            },
38            include: { madrasah: true },
39          });
40
41          if (!user || !user.isActive) {
42            return null;
43          }
44
45          const isValid = await bcrypt.compare(
46            parsed.data.password,
47            user.passwordHash
48          );
49
50          if (!isValid) {
51            return null;
52          }
53
54          return {
55            id: user.id,
56            name: user.name,
57            email: user.email,
```

```
58     role: user.role,
59     nip: user.nip,
60     madrasahId: user.madrasahId,
61     madrasahName: user.madrasah?.name ?? null,
62   };
63 },
64 )),
65 ],
66 callbacks: {
67   async jwt({ token, user }) {
68     if (user) {
69       token.id = user.id;
70       token.role = (user as unknown as { role: string }).role;
71       token.nip = (user as unknown as { nip: string | null }).nip;
72       token.madrasahId = (user as unknown as { madrasahId: string | null }).madrasah...
73       token.madrasahName = (user as unknown as { madrasahName: string | null }).madr...
74     }
75     return token;
76   },
77   async session({ session, token }) {
78     if (session.user) {
79       session.user.id = token.id as string;
80       (session.user as unknown as { role: string }).role = token.role as string;
81       (session.user as unknown as { nip: string | null }).nip = (token.nip as string...
82       (session.user as unknown as { madrasahId: string | null }).madrasahId =
83         token.madrasahId as string | null;
84       (session.user as unknown as { madrasahName: string | null }).madrasahName =
85         token.madrasahName as string | null;
86     }
87     return session;
88   },
89   async redirect({ url, baseUrl }) {
90     if (url.startsWith("/")) {
91       const resolvedBase = process.env.VERCEL_URL
92         ? `https://${process.env.VERCEL_URL}`
93         : process.env.NEXTAUTH_URL && !process.env.NEXTAUTH_URL.includes("localhost"...
94         ? process.env.NEXTAUTH_URL
95         : baseUrl;
96       return `${resolvedBase}${url}`;
97     }
98     try {
99       if (new URL(url).origin === baseUrl) return url;
100     } catch {}
101     return baseUrl;
102   },
103 },
104 pages: {
105   signIn: "/login",
106 },
107 session: {
108   strategy: "jwt",
109 },
110 secret: process.env.NEXTAUTH_SECRET,
111 });
```

4.2 Middleware Proteksi Rute (src/middleware.ts)

Berikut adalah kode sumber utuh `src/middleware.ts` untuk memisahkan lalu lintas akses Web Admin dan Portal Mobile Guru secara ketat:

```
KODE LENGKAP: src/middleware.ts
1  import { NextResponse } from "next/server";
2  import type { NextRequest } from "next/server";
3
4  export function middleware(request: NextRequest) {
5    const { pathname, searchParams } = request.nextUrl;
6
7    // Session cookie detection (compatible with both http and https NextAuth v5 cookies...
8    const sessionToken =
9      request.cookies.get("authjs.session-token")?.value ||
10     request.cookies.get("__Secure-authjs.session-token")?.value ||
11     request.cookies.get("next-auth.session-token")?.value ||
12     request.cookies.get("__Secure-next-auth.session-token")?.value;
```

```
13
14   const isAuthPage = pathname === "/login";
15   const isGuruAuthPage = pathname === "/guru/login";
16   const isProtectedAdminPage =
17     pathname.startsWith("/admin") ||
18     pathname.startsWith("/superadmin");
19   const isProtectedGuruPage =
20     pathname.startsWith("/guru") && !pathname.startsWith("/guru/login");
21
22   // If this is a Server Action request, bypass page-level redirect so Next.js receive...
23   // Server Actions handle authentication and return structured error messages via req...
24   if (request.headers.has("next-action")) {
25     return NextResponse.next();
26   }
27
28   // Developer testing override via query param ?nip=
29   if (searchParams.has("nip")) {
30     return NextResponse.next();
31   }
32
33   // If visiting protected admin/superadmin page without any session cookie, redirect ...
34   if (isProtectedAdminPage && !sessionToken) {
35     const loginUrl = new URL("/login", request.url);
36     loginUrl.searchParams.set("callbackUrl", pathname);
37     return NextResponse.redirect(loginUrl);
38   }
39
40   // If visiting protected guru mobile page without any session cookie, redirect to /g...
41   if (isProtectedGuruPage && !sessionToken) {
42     const loginUrl = new URL("/guru/login", request.url);
43     loginUrl.searchParams.set("callbackUrl", pathname);
44     return NextResponse.redirect(loginUrl);
45   }
46
47   // If already logged in and visiting web admin /login, redirect to /admin
48   if (isAuthPage && sessionToken) {
49     return NextResponse.redirect(new URL("/admin", request.url));
50   }
51
52   // If already logged in and visiting /guru/login, redirect to /guru
53   if (isGuruAuthPage && sessionToken) {
54     return NextResponse.redirect(new URL("/guru", request.url));
55   }
56
57   return NextResponse.next();
58 }
59
60 export const config = {
61   matcher: [
62     "/admin/:path*",
63     "/superadmin/:path*",
64     "/login",
65     "/guru/:path*",
66     "/guru",
67   ],
68 };

```

4.3 Lapisan Keamanan Server Action (src/server/utills/auth-guard.ts)

Mencegah celah kerentanan BOLA/IDOR (Broken Object Level Authorization) agar Admin Madrasah A tidak dapat memanipulasi data Madrasah B:

```
KODE LENCARA: src/server/utills/auth_guard.ts
1  import { auth } from "@lib/auth";
2  import { prisma } from "@lib/prisma";
3
4  export interface AuthenticatedUser {
5    id: string;
6    name?: string | null;
7    email?: string | null;
8    role: "SUPERADMIN" | "ADMIN_MADRASAH" | "TEACHER";
9    madrasahId?: string | null;
10   madrasahName?: string | null;

```

```
11     nip?: string | null;
12   }
13
14   /**
15    * Ensures user is authenticated. Throws error if unauthenticated.
16    */
17   export async function requireAuth(): Promise<AuthenticatedUser> {
18     const session = await auth();
19     const user = session?.user as unknown as AuthenticatedUser | undefined;
20
21     if (!user || !user.id) {
22       throw new Error("Sesi tidak valid atau telah berakhir. Silakan login kembali.");
23     }
24
25     return user;
26   }
27
28   /**
29    * Ensures user has SUPERADMIN role. Throws error if unauthorized.
30    */
31   export async function requireSuperadmin(): Promise<AuthenticatedUser> {
32     const user = await requireAuth();
33
34     if (user.role !== "SUPERADMIN") {
35       throw new Error("Akses ditolak: Diperlukan hak akses Superadmin.");
36     }
37
38     return user;
39   }
40
41   /**
42    * Ensures user has Administrator permissions (ADMIN_MADRASAH or SUPERADMIN)
43    * and strictly prevents cross-tenant access (BOLA/IDOR protection).
44    */
45   export async function requireMadrasahAdmin(
46     requestedMadrasahId?: string
47   ): Promise<{ user: AuthenticatedUser; madrasahId: string }> {
48     const user = await requireAuth();
49
50     if (user.role === "SUPERADMIN") {
51       if (requestedMadrasahId && requestedMadrasahId.trim().length > 0) {
52         return { user, madrasahId: requestedMadrasahId.trim() };
53       }
54       if (user.madrasahId) {
55         return { user, madrasahId: user.madrasahId };
56       }
57       const firstMadrasah = await prisma.madrasah.findFirst({
58         where: { isActive: true },
59         select: { id: true },
60       });
61       if (!firstMadrasah) {
62         throw new Error("Tidak ada data madrasah aktif di sistem.");
63       }
64       return { user, madrasahId: firstMadrasah.id };
65     }
66
67     if (user.role !== "ADMIN_MADRASAH") {
68       throw new Error("Akses ditolak: Diperlukan hak akses Administrator Madrasah.");
69     }
70
71     if (!user.madrasahId) {
72       throw new Error("Akun Administrator Anda belum terhubung ke madrasah manapun.");
73     }
74
75     // Cross-tenant access protection (IDOR / BOLA Prevention):
76     // An Admin of Madrasah A CANNOT access or mutate Madrasah B's data
77     if (
78       requestedMadrasahId &&
79       requestedMadrasahId.trim().length > 0 &&
80       requestedMadrasahId.trim() !== user.madrasahId
81     ) {
82       throw new Error("Akses ditolak: Anda tidak memiliki izin untuk mengelola data madr...");
83     }
84
85     return { user, madrasahId: user.madrasahId };
86   }
```

```
87
88  /**
89   * Resolves the authenticated user ID for attendance recording.
90   * Strictly prevents proxy attendance: Regular teachers CANNOT submit attendance on be...
91   */
92  export async function getEffectiveAttendanceUserId(
93    requestedUserId?: string
94  ): Promise<{ user: AuthenticatedUser; effectiveUserId: string }> {
95    const user = await requireAuth();
96
97    // If user is a teacher, they can ONLY record attendance for themselves
98    if (user.role === "TEACHER") {
99      return { user, effectiveUserId: user.id };
100    }
101
102    // If user is an Admin / Superadmin, they can specify a teacherId for manual input
103    if (requestedUserId && requestedUserId.trim().length > 0) {
104      return { user, effectiveUserId: requestedUserId.trim() };
105    }
106
107    return { user, effectiveUserId: user.id };
108  }
```

BAB 5

Core Modul 1: Geofencing GPS & Algoritma Haversine

Membedah formula matematika geodetik Haversine untuk menghitung jarak kurva bumi dalam meter, validasi koordinat GPS guru terhadap gerbang madrasah, dan kalkulasi toleransi keterlambatan.

5.1 Rumus Matematika Haversine

Jarak dua titik koordinat permukaan bumi (latitude dan longitude) dihitung menggunakan rumus Great-Circle Distance dengan konstanta radius bumi R = 6.371.000 meter. Berikut adalah kode sumber utuh implementasinya:

```
KODE LENGKAP: src/lib/geos.ts
1  /**
2   * Utility functions for Geofencing & Coordinates Calculation (WGS84)
3   */
4
5   export function calculateDistanceMeters(
6     lat1: number,
7     lon1: number,
8     lat2: number,
9     lon2: number
10  ): number {
11    const R = 6371000; // Earth radius in meters
12    const dLat = (lat2 - lat1) * (Math.PI / 180);
13    const dLon = (lon2 - lon1) * (Math.PI / 180);
14    const a =
15      Math.sin(dLat / 2) * Math.sin(dLat / 2) +
16      Math.cos(lat1 * (Math.PI / 180)) *
17      Math.cos(lat2 * (Math.PI / 180)) *
18      Math.sin(dLon / 2) *
19      Math.sin(dLon / 2);
20    const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
21    return Math.round(R * c * 10) / 10;
22  }
23
24  export function isWithinGeofence(
25    currentLat: number,
26    currentLng: number,
27    targetLat: number,
28    targetLng: number,
29    radiusMeters: number
30  ): { isInside: boolean; distance: number } {
31    const distance = calculateDistanceMeters(currentLat, currentLng, targetLat, targetLn...
32    return {
33      isInside: distance <= radiusMeters,
34      distance,
35    };
36  }
```

5.2 Validasi Jam Kerja & Toleransi Keterlambatan

Server memvalidasi jam saat transaksi presensi dikirim terhadap waktu jam mulai kerja (`workStartTime`) dan ambang batas terlambat (`lateThreshold`). Jika jam masuk melebihi batas toleransi, status otomatis tercatat sebagai `LATE` (Terlambat).

BAB 6

Core Modul 2: Swafoto Kamera Live & Watermarking Canvas

Membuka stream kamera perangkat, membakar metadata legalitas berupa nama, NIP, tanggal, jam WIB, dan koordinat GPS langsung ke atas frame foto menggunakan HTML5 2D Canvas.

6.1 Implementasi Modal Kamera Live & Watermarking Canvas

Berikut adalah kode sumber utuh komponen `mobile-camera-modal.tsx` yang menangani akses hardware kamera, tombol flip depan/belakang, serta penempelan watermark resmi:

```
KODE LENGKAP: src/features/mobile-app/components/mobile-camera-modal.tsx
1  "use client";
2
3  import React, { useState, useRef, useEffect, useCallback } from "react";
4  import { createPortal } from "react-dom";
5  import {
6    Camera,
7    RefreshCw,
8    X,
9    CheckCircle2,
10   MapPin,
11   Clock,
12   Sparkles,
13   ShieldCheck,
14   AlertCircle,
15   FlipHorizontal,
16   Upload
17 } from "lucide-react";
18 import { Button } from "@components/atoms/button";
19 import { Badge } from "@components/atoms/badge";
20
21 interface MobileCameraModalProps {
22   isOpen: boolean;
23   type: "CHECK_IN" | "CHECK_OUT";
24   teacherName: string;
25   teacherNip: string;
26   madrasahName: string;
27   currentLat: number;
28   currentLng: number;
29   distanceMeters: number;
30   isInsideRadius: boolean;
31   onClose: () => void;
32   onSubmit: (photoBase64: string, notes?: string) => Promise<void>;
33 }
34
35 export function MobileCameraModal({
36   isOpen,
37   type,
38   teacherName,
39   teacherNip,
40   madrasahName,
41   currentLat,
42   currentLng,
43   distanceMeters,
44   isInsideRadius,
45   onClose,
46   onSubmit,
47 }: MobileCameraModalProps) {
48   const [facingMode, setFacingMode] = useState<"user" | "environment">("user");
49   const [isStreaming, setIsStreaming] = useState(false);
50   const [cameraError, setCameraError] = useState<string | null>(null);
51   const [capturedPhoto, setCapturedPhoto] = useState<string | null>(null);
52   const [isSubmitting, setIsSubmitting] = useState(false);
53   const [notes, setNotes] = useState("");
54
55   const videoRef = useRef<HTMLVideoElement | null>(null);
56   const canvasRef = useRef<HTMLCanvasElement | null>(null);
57   const streamRef = useRef<MediaStream | null>(null);
```

```
58     const fileInputRef = useRef<HTMLInputElement | null>(null);
59
60     const [isMounted, setIsMounted] = useState(false);
61     useEffect(() => {
62         setIsMounted(true);
63     }, []);
64
65     // Stop current video stream
66     const stopStream = useCallback(() => {
67         if (streamRef.current) {
68             streamRef.current.getTracks().forEach((track) => track.stop());
69             streamRef.current = null;
70         }
71         setIsStreaming(false);
72     }, []);
73
74     // Start webcam stream
75     const startCamera = useCallback(async () => {
76         stopStream();
77         setCameraError(null);
78
79         try {
80             if (!navigator.mediaDevices || !navigator.mediaDevices.getUserMedia) {
81                 throw new Error("Kamera tidak didukung pada browser ini.");
82             }
83
84             const stream = await navigator.mediaDevices.getUserMedia({
85                 video: {
86                     facingMode: facingMode,
87                     width: { ideal: 720 },
88                     height: { ideal: 960 },
89                 },
90                 audio: false,
91             });
92
93             streamRef.current = stream;
94             if (videoRef.current) {
95                 videoRef.current.srcObject = stream;
96                 videoRef.current.onloadedmetadata = () => {
97                     videoRef.current?.play().catch(() => {});
98                     setIsStreaming(true);
99                 };
100             }
101         } catch (err: any) {
102             console.warn("Kamera tidak dapat diakses:", err);
103             setCameraError(
104                 err.name === "NotAllowedError" || err.name === "PermissionDeniedError"
105                 ? "Izin akses kamera ditolak. Silakan izinkan akses kamera atau gunakan simu..."
106                 : "Perangkat kamera tidak terdeteksi atau sedang digunakan aplikasi lain."
107             );
108         }
109     }, [facingMode, stopStream]);
110
111     useEffect(() => {
112         if (isOpen && !capturedPhoto) {
113             startCamera();
114         } else {
115             stopStream();
116         }
117
118         return () => {
119             stopStream();
120         };
121     }, [isOpen, capturedPhoto, startCamera, stopStream]);
122
123     // Flip Camera Front / Back
124     const toggleFacingMode = () => {
125         setFacingMode((prev) => (prev === "user" ? "environment" : "user"));
126     };
127
128     // Capture Snapshot and Burn Watermark
129     const handleCapture = () => {
130         if (!videoRef.current || !canvasRef.current) return;
131
132         const video = videoRef.current;
133         const canvas = canvasRef.current;
```

```
134     const ctx = canvas.getContext("2d");
135     if (!ctx) return;
136
137     const w = video.videoWidth || 640;
138     const h = video.videoHeight || 480;
139
140     canvas.width = w;
141     canvas.height = h;
142
143     // Draw video frame (mirror if front camera)
144     ctx.save();
145     if (facingMode === "user") {
146         ctx.translate(w, 0);
147         ctx.scale(-1, 1);
148     }
149     ctx.drawImage(video, 0, 0, w, h);
150     ctx.restore();
151
152     // Burn Official Watermark
153     const barHeight = Math.max(80, Math.floor(h * 0.16));
154     const gradient = ctx.createLinearGradient(0, h - barHeight - 20, 0, h);
155     gradient.addColorStop(0, "rgba(0, 0, 0, 0)");
156     gradient.addColorStop(0.3, "rgba(5, 96, 58, 0.75)");
157     gradient.addColorStop(1, "rgba(5, 96, 58, 0.95)");
158
159     ctx.fillStyle = gradient;
160     ctx.fillRect(0, h - barHeight - 20, w, barHeight + 20);
161
162     // Watermark Texts
163     ctx.fillStyle = "#FFFFFF";
164     ctx.font = `bold ${Math.floor(w * 0.034)}px system-ui, sans-serif`;
165     ctx.fillText(`${teacherName} (${teacherNip})`, 20, h - barHeight + 10);
166
167     ctx.font = `${Math.floor(w * 0.026)}px system-ui, sans-serif`;
168     ctx.fillStyle = "#E0E7FF";
169     ctx.fillText(madrasahName, 20, h - barHeight + 30);
170
171     const now = new Date();
172     const timeStr = now.toLocaleDateString("id-ID", {
173         day: "2-digit",
174         month: "short",
175         year: "numeric",
176         hour: "2-digit",
177         minute: "2-digit",
178         second: "2-digit",
179     }) + " WIB";
180
181     ctx.fillStyle = "#FDE047"; // Yellow accent
182     ctx.font = `600 ${Math.floor(w * 0.025)}px system-ui, sans-serif`;
183     ctx.fillText(
184         `#np    ${timeStr}    • GPS:    ${currentLat.toFixed(5)},    ${currentLat.toFixed(5)},
185         20,
186         h - barHeight + 50
187     );
188
189     const dataUrl = canvas.toDataURL("image/jpeg", 0.85);
190     setCapturedPhoto(dataUrl);
191     stopStream();
192 };
193
194 // Fallback: Upload Photo / Simulation Photo
195 const handleFileUpload = (e: React.ChangeEvent<HTMLInputElement>) => {
196     const file = e.target.files?.[0];
197     if (!file) return;
198
199     const reader = new FileReader();
200     reader.onload = (event) => {
201         const img = new Image();
202         img.onload = () => {
203             const canvas = canvasRef.current;
204             if (!canvas) return;
205             const ctx = canvas.getContext("2d");
206             if (!ctx) return;
207
208             canvas.width = img.width;
209             canvas.height = img.height;
```

```
210      ctx.drawImage(img, 0, 0);
211
212      // Watermark
213      const barHeight = Math.max(80, Math.floor(img.height * 0.16));
214      ctx.fillStyle = "rgba(5, 96, 58, 0.85)";
215      ctx.fillRect(0, img.height - barHeight, img.width, barHeight);
216
217      ctx.fillStyle = "#FFFFFF";
218      ctx.font = `bold ${Math.floor(img.width * 0.035)}px system-ui, sans-serif`;
219      ctx.fillText(`${teacherName} • ${teacherNip}`, 20, img.height - barHeight + 30...
220
221      ctx.font = `${Math.floor(img.width * 0.026)}px system-ui, sans-serif`;
222      ctx.fillStyle = "#FDE047";
223      ctx.fillText(
224        `#ñp    $ { new    Date ( ) . to Local e Time String ( " i d - I D " ) }    W I B
225        20,
226        img.height - barHeight + 58
227      );
228
229      setCapturedPhoto(canvas.toDataURL("image/jpeg", 0.85));
230      stopStream();
231    };
232    img.src = event.target?.result as string;
233  };
234  reader.readAsDataURL(file);
235  };
236
237  // Submit Photo
238  const handleSendAttendance = async () => {
239    if (!capturedPhoto) return;
240    setIsSubmitting(true);
241    try {
242      await onSubmit(capturedPhoto, notes);
243      onClose();
244    } finally {
245      setIsSubmitting(false);
246    }
247  };
248
249  if (!isOpen || !isMounted) return null;
250
251  return createPortal(
252    <div className="fixed inset-0 z-[999] flex items-center justify-center bg-black/85...
253      {/} Hidden canvas for snapshot watermark rendering */}
254      <canvas ref={canvasRef} className="hidden" />
255      <input
256        ref={fileInputRef}
257        type="file"
258        accept="image/*"
259        capture="user"
260        className="hidden"
261        onChange={handleFileUpload}
262      />
263
264      <div className="relative w-full max-w-md bg-slate-900 border border-slate-800 ro...
265        {/} Top Header Bar */}
266        <div className="flex items-center justify-between px-4 py-3 bg-gradient-to-r f...
267          <div className="flex items-center gap-2">
268            <div className="w-8 h-8 rounded-full bg-emerald-500/20 flex items-center j...
269              <Camera className="w-4 h-4 text-emerald-300" />
270            </div>
271            <div>
272              <h3 className="text-sm font-bold leading-tight">
273                {type === "CHECK_IN" ? "Selfie Presensi Masuk" : "Selfie Presensi Pula...
274              </h3>
275              <p className="text-[11px] text-emerald-200">Verifikasi Wajah & Geofence<...
276            </div>
277          </div>
278          <button
279            onClick={onClose}
280            className="w-8 h-8 rounded-full bg-white/10 hover:bg-white/20 flex items-c...
281          >
282            <X className="w-4 h-4" />
283          </button>
284        </div>
285
```

```
286      { /* Viewfinder Area */ }
287      <div className="relative flex-1 bg-black flex items-center justify-center over...
288      { capturedPhoto ? (
289        /* Snapshot Review View */
290        <div className="relative w-full h-full flex flex-col items-center justify-...
291        { /* eslint-disable-next-line @next/next/no-img-element */ }
292        <img
293          src={capturedPhoto}
294          alt="Selfie Presensi"
295          className="w-full h-auto max-h-[380px] object-contain rounded-lg"
296        />
297        <div className="absolute top-3 left-3 bg-emerald-600/90 text-white text-...
298        <CheckCircle2 className="w-3.5 h-3.5" />
299        Foto Berhasil Diambil
300      </div>
301    </div>
302  ) : (
303    /* Live Camera Stream */
304    <div className="relative w-full h-full flex items-center justify-center">
305      { cameraError ? (
306        <div className="p-6 text-center text-slate-300 flex flex-col items-cen...
307        <div className="w-14 h-14 rounded-full bg-amber-500/20 text-amber-40...
308        <AlertCircle className="w-7 h-7" />
309      </div>
310      <p className="text-xs text-amber-200 leading-relaxed max-w-xs">{came...
311      <div className="flex flex-col gap-2 w-full max-w-xs mt-2">
312        <Button
313          size="sm"
314          onClick={startCamera}
315          className="bg-emerald-600 hover:bg-emerald-500 text-white rounde...
316        >
317        <RefreshCw className="w-4 h-4" />
318        Coba Akses Kamera Lagi
319      </Button>
320      <Button
321        size="sm"
322        variant="outline"
323        onClick={() => fileInputRef.current?.click()}
324        className="border-slate-700 text-slate-200 hover:bg-slate-800 ro...
325      >
326      <Upload className="w-4 h-4" />
327      Pilih / Ambil Foto File
328    </Button>
329  </div>
330 </div>
331 ) : (
332   <>
333     <video
334       ref={videoRef}
335       autoPlay
336       playsInline
337       muted
338       className={`w-full h-full object-cover max-h-[420px] ${
339         facingMode === "user" ? "-scale-x-100" : ""
340       }`}
341     />
342
343     { /* Face Guide Oval */ }
344     <div className="absolute inset-0 pointer-events-none flex items-cent...
345     <div className="w-56 h-72 border-2 border-dashed border-emerald-40...
346     <div className="text-center text-emerald-200/80 text-[11px] bg-b...
347     Posisikan Wajah di Sini
348   </div>
349 </div>
350 </div>
351
352 { /* Distance & Geofence Floating Badge */ }
353 <div className="absolute top-3 left-3 right-3 flex items-center just...
354 <div
355   className={`text-[11px] font-medium px-2.5 py-1 rounded-full bac...
356   isInsideRadius
357   ? "bg-emerald-600/90 text-white border border-emerald-400/40...
358   : "bg-rose-600/90 text-white border border-rose-400/40"
359 }`}
360 >
361 <MapPin className="w-3.5 h-3.5" />
```

```
362         <span>{distanceMeters.toFixed(1)}m</span>
363         <span>•</span>
364         <span>{isInsideRadius ? "Dalam Radius" : "Di Luar Radius"}</span...
365     </div>
366
367     <button
368         type="button"
369         onClick={toggleFacingMode}
370         className="pointer-events-auto w-8 h-8 rounded-full bg-black/50 ...
371         title="Balik Kamera Depan/Belakang"
372     >
373         <FlipHorizontal className="w-4 h-4" />
374     </button>
375 </div>
376 </>
377 }}
378 </div>
379 }}
380 </div>
381
382 /* Bottom Action Controls */
383 <div className="p-4 bg-slate-900 border-t border-slate-800 flex flex-col gap-3...
384 {capturedPhoto ? (
385     /* Review and Confirm Controls */
386     <div className="flex flex-col gap-2.5">
387         <input
388             type="text"
389             placeholder="Catatan kehadiran (opsional, misal: Piket gerbang)..."
390             value={notes}
391             onChange={(e) => setNotes(e.target.value)}
392             className="w-full text-xs px-3 py-2 bg-slate-800 border border-slate-7...
393         />
394
395         <div className="flex gap-2">
396             <Button
397                 type="button"
398                 variant="outline"
399                 onClick={() => {
400                     setCapturedPhoto(null);
401                     startCamera();
402                 }}
403                 className="flex-1 border-slate-700 text-slate-300 hover:bg-slate-800...
404             >
405                 <RefreshCw className="w-3.5 h-3.5 mr-1" />
406                 Foto Ulang
407             </Button>
408
409             <Button
410                 type="button"
411                 onClick={handleSendAttendance}
412                 disabled={isSubmitting}
413                 className="flex-1 bg-gradient-to-r from-emerald-600 to-teal-600 hove...
414             >
415                 {isSubmitting ? (
416                     <span className="flex items-center gap-1.5">
417                         <RefreshCw className="w-3.5 h-3.5 animate-spin" />
418                         Mengirim...
419                     </span>
420                 ) : (
421                     <span className="flex items-center gap-1.5">
422                         <CheckCircle2 className="w-3.5 h-3.5" />
423                         Kirim Presensi Sekarang
424                     </span>
425                 )}
426             </Button>
427         </div>
428     </div>
429 ) : (
430     /* Capture Controls */
431     <div className="flex items-center justify-around py-1">
432         <button
433             type="button"
434             onClick={() => fileInputRef.current?.click()}
435             className="flex flex-col items-center gap-1 text-slate-400 hover:text-...
436         >
437         <div className="w-10 h-10 rounded-full bg-slate-800 flex items-center ...
```

```
438         <Upload className="w-4 h-4" />
439     </div>
440     <span className="text-[10px]">Unggah</span>
441 </button>
442
443     {/ * Shutter Button (Flutter Style) */}
444     <button
445         type="button"
446         onClick={handleCapture}
447         disabled={!isStreaming && !cameraError}
448         className="w-16 h-16 rounded-full border-4 border-white flex items-cen...
449     >
450         <div className="w-full h-full rounded-full bg-emerald-500 hover:bg-eme...
451             <Camera className="w-6 h-6 text-white" />
452         </div>
453     </button>
454
455     <button
456         type="button"
457         onClick={toggleFacingMode}
458         className="flex flex-col items-center gap-1 text-slate-400 hover:text-...
459     >
460         <div className="w-10 h-10 rounded-full bg-slate-800 flex items-center ...
461             <FlipHorizontal className="w-4 h-4" />
462         </div>
463         <span className="text-[10px]">Balik</span>
464     </button>
465 </div>
466 }}
467 </div>
468 </div>
469 </div>,
470 document.body
471 );
472 }
```

6.2 Server Action Perekaman Presensi (recordMobileAttendanceAction)

Fungsi backend yang menerima koordinat GPS dan string foto Base64 untuk divalidasi geofence sebelum disimpan ke database:

```
KODE LENGKAP: src/server/actions/mobile-attendance-actions.ts (Bagian 1: Inisialisasi & Query)
1  "use server";
2
3  import { prisma } from "@lib/prisma";
4  import { auth } from "@lib/auth";
5  import { isWithinGeofence } from "@lib/geo";
6  import { revalidatePath } from "next/cache";
7  import { AttendanceStatus } from "@prisma/client";
8
9  function safeRevalidatePath(path: string) {
10    try {
11      revalidatePath(path);
12    } catch {
13      // Ignore when called outside of Next.js request context (e.g. testing)
14    }
15  }
16
17  export interface MobileAttendancePayload {
18    userId?: string;
19    type: "CHECK_IN" | "CHECK_OUT" | "FULL";
20    lat?: number;
21    lng?: number;
22    distance?: number;
23    photoBase64?: string;
24    notes?: string;
25    dateStr?: string; // YYYY-MM-DD for backdated/retroactive attendance
26    customCheckInTime?: string; // HH:mm
27    customCheckOutTime?: string; // HH:mm
28  }
29
30  /**
31   * Bulletproof helper to save attendanceLog without Unique constraint failed on (user_...
```

```
32  */
33  async function upsertAttendanceRecord({
34    logId,
35    madrasahId,
36    userId,
37    date,
38    data,
39  }: {
40    logId?: string | null;
41    madrasahId: string;
42    userId: string;
43    date: Date;
44    data: any;
45  }) {
46    if (logId) {
47      try {
48        return await prisma.attendanceLog.update({
49          where: { id: logId },
50          data,
51        });
52      } catch {
53        // Fall through to upsert
54      }
55    }
56
57    try {
58      return await prisma.attendanceLog.upsert({
59        where: {
60          userId_date: {
61            userId,
62            date,
63          },
64        },
65        create: {
66          madrasahId,
67          userId,
68          date,
69          ...data,
70        },
71        update: data,
72      });
73    } catch (err: any) {
74      const dStart = new Date(date.getFullYear(), date.getMonth(), date.getDate() - 1, 0...
75      const dEnd = new Date(date.getFullYear(), date.getMonth(), date.getDate() + 1, 23,...
76
77      const existing = await prisma.attendanceLog.findFirst({
78        where: {
79          userId,
80          date: { gte: dStart, lte: dEnd },
81        },
82        orderBy: { createdAt: "desc" },
83      });
84
85      if (existing) {
86        return await prisma.attendanceLog.update({
87          where: { id: existing.id },
88          data,
89        });
90      }
91
92      throw err;
93    }
94  }
95
96  /**
97   * Fetch teacher profile, madrasah geofence settings, today's attendance status, and h...
98   */
99  export async function getTeacherMobileDashboardData(explicitUserId?: string) {
100    try {
101      let userId = explicitUserId;
102
103      if (!userId) {
104        const session = await auth();
105        userId = session?.user?.id;
106      }
107    }
```

```
108     if (!userId) {
109         return { error: "Belum login atau sesi telah berakhir." };
110     }
111
112     let teacher: any = null;
113     try {
114         teacher = await prisma.user.findUnique({
115             where: { id: userId },
116             include: {
117                 position: true,
118                 madrasah: {
119                     include: {
120                         settings: true,
121                     },
122                 },
123             },
124         });
125     } catch {
126         teacher = await prisma.user.findUnique({
127             where: { id: userId },
128             include: {
129                 madrasah: {
130                     include: {
131                         settings: true,
132                     },
133                 },
134             },
135         });
136     }
137
138     if (!teacher) {
139         return { error: "Data pengguna tidak ditemukan." };
140     }
141
142     // If the logged-in user is an ADMIN / SUPERADMIN testing the mobile portal:
143     if (teacher.role !== "TEACHER") {
144         const targetMadrasahId = teacher.madrasahId;
145         let previewTeacher: any = null;
146
147         if (targetMadrasahId) {
148             try {
149                 previewTeacher = await prisma.user.findFirst({
150                     where: { madrasahId: targetMadrasahId, role: "TEACHER", isActive: true },
151                     include: {
152                         position: true,
153                         madrasah: {
154                             include: { settings: true },
155                         },
156                     },
157                     orderBy: { name: "asc" },
158                 });
159             } catch {
160                 previewTeacher = await prisma.user.findFirst({
161                     where: { madrasahId: targetMadrasahId, role: "TEACHER", isActive: true },
162                     include: {
163                         madrasah: {
164                             include: { settings: true },
165                         },
166                     },
167                     orderBy: { name: "asc" },
168                 });
169             }
170         }
171
172         if (!previewTeacher && teacher.role === "SUPERADMIN") {
173             try {
174                 previewTeacher = await prisma.user.findFirst({
175                     where: { role: "TEACHER", isActive: true },
176                     include: {
177                         position: true,
178                         madrasah: {
179                             include: { settings: true },
180                         },
181                     },
182                     orderBy: { name: "asc" },
183                 });
```

```
184     } catch {
185       previewTeacher = await prisma.user.findFirst({
186         where: { role: "TEACHER", isActive: true },
187         include: {
188           madrasah: {
189             include: { settings: true },
190           },
191         },
192         orderBy: { name: "asc" },
193       });
194     }
195   }
196
197   if (previewTeacher) {
198     teacher = previewTeacher;
199   }
200 }
201
202 if (!teacher || !teacher.madrasah) {
203   return { error: "Data guru atau madrasah tidak ditemukan." };
204 }
205
206 const madrasahId = teacher.madrasah.id;
207 const settings = teacher.madrasah.settings ?? {
208   latitude: -6.2615,
209   longitude: 106.8106,
210   radiusMeters: 50,
211   workStartTime: "07:00",
212   lateThreshold: "07:15",
213   workEndTime: "14:00",
214   workDays: "1,2,3,4,5,6",
215   requireSelfie: true,
216 };
217
218 // Calculate today's midnight UTC and range
219 const now = new Date();
220 const todayStart = new Date(now.getFullYear(), now.getMonth(), now.getDate(), 0, 0...
221 const todayEnd = new Date(now.getFullYear(), now.getMonth(), now.getDate(), 23, 59...
222
223 // 1. Fetch Today's Attendance Log
224 const todayLog = await prisma.attendanceLog.findFirst({
225   where: {
226     userId: teacher.id,
227     madrasahId,
228     date: {
229       gte: todayStart,
230       lte: todayEnd,
231     },
232   },
233 });
234
235 // 2. Check if Today is Holiday (Single or Multi-day / Semester break)
236 const holiday = await prisma.holiday.findFirst({
237   where: {
238     madrasahId,
239     OR: [
240       {
241         date: {
242           gte: todayStart,
243           lte: todayEnd,
244         },
245       },
246       {
247         date: { lte: todayEnd },
248         endDate: { gte: todayStart },
249       },
250     ],
251   },
252 });
253
254 // 3. Check day of week (0 = Sunday, 1 = Monday, etc.)
255 const dayOfWeek = now.getDay().toString();
256 const activeWorkDays = (settings.workDays || "1,2,3,4,5,6").split(",");
257 const isWeekend = !activeWorkDays.includes(dayOfWeek);
258
259 // 4. Calculate real monthly statistics for this teacher from database
```

```
260     const startOfMonth = new Date(now.getFullYear(), now.getMonth(), 1, 0, 0, 0, 0);
261     const endOfMonth = new Date(now.getFullYear(), now.getMonth() + 1, 0, 23, 59, 59, ...
262
263     const monthlyLogs = await prisma.attendanceLog.findMany({
264       where: {
265         userId: teacher.id,
266         madrasahId,
267         date: {
268           gte: startOfMonth,
269           lte: endOfMonth,
270         },
271       },
272     });
273
274     const presentMonth = monthlyLogs.filter((l) => l.status === "PRESENT").length;
275     const lateMonth = monthlyLogs.filter((l) => l.status === "LATE").length;
276     const permitMonth = monthlyLogs.filter((l) => l.status === "PERMIT").length;
277     const sickMonth = monthlyLogs.filter((l) => l.status === "SICK").length;
278     const absentMonth = monthlyLogs.filter((l) => l.status === "ABSENT").length;
279     const totalMonth = monthlyLogs.length;
280
281     let totalWorkMinutes = 0;
282     monthlyLogs.forEach((l) => {
283       if (l.
```

BAB 7

Core Modul 3: Antarmuka Mobile GTK & Progressive Web App (PWA)

Membangun pengalaman aplikasi mobile native di dalam peramban web (Progressive Web App) dengan navigasi Material 3, install prompt, status offline, dan Service Worker.

7.1 Kode Sumber Shell Navigasi Mobile (MobileAppShell)

Shell navigasi dengan 5 tab utama (Beranda, Jurnal KBM, Riwayat, Izin, dan Profil Pengguna):

```
KODE LENGKAP: src/features/mobile-app/components/mobile-app-shell.tsx
1  "use client";
2
3  import React, { useState } from "react";
4  import { MobileHomeView } from "../mobile-home-view";
5  import { MobileHistoryView } from "../mobile-history-view";
6  import { MobilePermitView } from "../mobile-permit-view";
7  import { MobileProfileView } from "../mobile-profile-view";
8  import { MobileJournalView } from "../mobile-journal-view";
9
10 interface MobileAppShellProps {
11   data: {
12     teacher: {
13       id: string;
14       name: string;
15       nip: string;
16       email: string;
17       phone: string;
18       avatarUrl?: string | null;
19       madrasahName: string;
20       madrasahAddress: string;
21     };
22     settings: {
23       latitude: number;
24       longitude: number;
25       radiusMeters: number;
26       workStartTime: string;
27       lateThreshold: string;
28       workEndTime: string;
29       requireSelfie: boolean;
30       allowBackdatedAttendance?: boolean;
31     };
32     todayLog?: {
33       id: string;
34       status: string;
35       checkInTime: string | null;
36       checkInDistance: number | null;
37       checkInPhotoUrl: string | null;
38       checkInLat?: number | null;
39       checkInLng?: number | null;
40       checkOutTime: string | null;
41       checkOutDistance: number | null;
42       checkOutPhotoUrl: string | null;
43       notes?: string | null;
44     } | null;
45     holiday?: {
46       name: string;
47       isNational: boolean;
48       description?: string | null;
49     } | null;
50     isWeekend?: boolean;
51   };
52   onRefresh: () => void;
53   initialTab?: TabId;
54 }
55
56 type TabId = "home" | "journal" | "history" | "permit" | "profile";
57
58 const tabs: { id: TabId; label: string; icon: React.ReactNode }[] = [
```

```
59      {
60        id: "home",
61        label: "Beranda",
62        icon: (
63          <svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="currentColor...
64            <path d="M3 9l9-7 9 7v11a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>
65            <polyline points="9 22 9 12 15 12 15 22"/>
66          </svg>
67        ),
68      },
69      {
70        id: "journal",
71        label: "Jurnal KBM",
72        icon: (
73          <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor...
74            <path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/>
75            <path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/>
76            <line x1="8" y1="7" x2="16" y2="7"/>
77            <line x1="8" y1="11" x2="14" y2="11"/>
78          </svg>
79        ),
80      },
81      {
82        id: "history",
83        label: "Riwayat",
84        icon: (
85          <svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="currentColor...
86            <path d="M3 3h6l3 18 3-12 2 6h4"/>
87          </svg>
88        ),
89      },
90      {
91        id: "permit",
92        label: "Izin",
93        icon: (
94          <svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="currentColor...
95            <path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/>
96            <polyline points="14 2 14 8 20 8"/>
97            <line x1="16" y1="13" x2="8" y2="13"/>
98            <line x1="16" y1="17" x2="8" y2="17"/>
99            <polyline points="10 9 9 8 9"/>
100          </svg>
101        ),
102      },
103      {
104        id: "profile",
105        label: "Profil",
106        icon: (
107          <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor...
108            <path d="M20 21v-2a4 4 0 0 0-4-4H8a4 4 0 0 0-4 4v2"/>
109            <circle cx="12" cy="7" r="4"/>
110          </svg>
111        ),
112      },
113    ];
114
115    const tabOrder: TabId[] = ["home", "journal", "history", "permit", "profile"];
116
117    export function MobileAppShell({ data, onRefresh, initialTab }: MobileAppShellProps) {
118      const [activeTab, setActiveTab] = useState<TabId>(() => {
119        if (initialTab && tabOrder.includes(initialTab)) return initialTab;
120        return "home";
121      });
122      const [animationClass, setAnimationClass] = useState<string>("animate-tab-fade-up");
123
124      const handleTabChange = (nextTab: TabId) => {
125        if (nextTab === activeTab) return;
126        const currentIndex = tabOrder.indexOf(activeTab);
127        const nextIndex = tabOrder.indexOf(nextTab);
128        setAnimationClass(nextIndex > currentIndex ? "animate-tab-slide-right" : "animate-...
129        setActiveTab(nextTab);
130      };
131
132      return (
133        <div
134          className="relative w-full min-h-screen flex flex-col text-[#0b1c30] select-none...
```

```
135     style={{
136       fontFamily: "'Plus Jakarta Sans', sans-serif",
137       backgroundColor: "#f8f9ff",
138     }}
139   >
140     { / * % %   T O P   H E A D E R   % %   * / }
141   <header
142     className="fixed top-0 w-full z-50 pt-safe"
143     style={{
144       background: "rgba(248,249,255,0.88)",
145       backdropFilter: "blur(16px)",
146       WebkitBackdropFilter: "blur(16px)",
147       boxShadow: "0 1px 8px rgba(0,0,0,0.04)",
148       maxWidth: "inherit",
149     }}
150   >
151   <div className="h-16 px-4 flex items-center justify-between">
152     {/ * Logo + Title with Animated Crossfade */}
153     <div className="flex items-center gap-2">
154       {/ * eslint-disable-next-line @next/next/no-img-element */}
155        { (e.target as HTMLImageElement).style.display = "none";...
160       />
161       <div key={activeTab} className="flex flex-col animate-title-fade">
162         <span className="text-[11px] text-[#444653] font-semibold leading-none">...
163         <h1 className="text-[14px] font-bold text-[#0b1c30] leading-tight">
164           {activeTab === "home" && "Beranda"}
165           {activeTab === "journal" && "Jurnal Pembelajaran KBM"}
166           {activeTab === "history" && "Riwayat Kehadiran"}
167           {activeTab === "permit" && "Pengajuan Izin"}
168           {activeTab === "profile" && "Profil Pengguna"}
169         </h1>
170       </div>
171     </div>
172
173     {/ * Right actions */}
174     <div className="flex items-center gap-1">
175       {/ * Notif bell */}
176       <button
177         aria-label="Notifikasi"
178         className="w-11 h-11 relative flex items-center justify-center rounded-f...
179       >
180         <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="curr...
181         <path d="M18 8A6 6 0 0 0 6 8c0 7-3 9-3 9h18s-3-2-3-9"/>
182         <path d="M13.73 21a2 2 0 0 1-3.46 0"/>
183       </svg>
184       <span className="absolute top-2.5 right-2.5 w-2 h-2 rounded-full bg-[#ba...
185     </button>
186     {/ * Avatar Button */}
187     <button
188       type="button"
189       onClick={() => handleTabChange("profile")}
190       className={`w-8 h-8 rounded-full flex items-center justify-center shrink...
191       activeTab === "profile" ? "ring-2 ring-[#00288e] bg-[#00288e]" : "bg-[...
192     `}
193     title="Profil Guru"
194   >
195     <span className="text-white text-[13px] font-bold">
196       {data.teacher.name.charAt(0)}
197     </span>
198   </button>
199 </div>
200 </div>
201 </header>
202
203     { / * % %   S C R O L L A B L E   B O D Y   W I T H   D I R E C T I O N A L   A N I M A T I O N   % %   *
204   <main
205     className="flex-1 overflow-y-auto"
206     style={{ padding: "64px 0 80px 0" }}
207   >
208   <div key={activeTab} className={`px-4 py-4 ${animationClass}`}>
209     {activeTab === "home" && (
210       <MobileHomeView
```

```
211      data={data}
212      onRefresh={onRefresh}
213      onOpenHistoryTab={() => handleTabChange("permit")}
214      onOpenJournalTab={() => handleTabChange("journal")}
215    />
216  })
217  {activeTab === "journal" && (
218    <MobileJournalView userId={data.teacher.id} />
219  )}
220  {activeTab === "history" && (
221    <MobileHistoryView userId={data.teacher.id} />
222  )}
223  {activeTab === "permit" && (
224    <MobilePermitView
225      userId={data.teacher.id}
226      onSuccess={() => {
227        onRefresh();
228        handleTabChange("history");
229      }}
230    />
231  )}
232  {activeTab === "profile" && (
233    <MobileProfileView
234      teacher={data.teacher}
235      settings={data.settings}
236    />
237  )}
238 </div>
239 </main>
240
241 { / * % %   B O T T O M   N A V   B A R   ( M a t e r i a l   3   s t y l e ) % %   * / }
242 <nav
243   className="fixed bottom-0 w-full z-50 pb-safe"
244   style={{
245     background: "rgba(255,255,255,0.92)",
246     backdropFilter: "blur(20px)",
247     WebkitBackdropFilter: "blur(20px)",
248     boxShadow: "0 -4px 16px -2px rgba(15,23,42,0.05)",
249     maxWidth: "inherit",
250   }}
251 >
252   <div className="h-16 px-4 flex items-center justify-around">
253     {tabs.map((tab) => {
254       const isActive = activeTab === tab.id;
255
256       return (
257         <button
258           key={tab.id}
259           type="button"
260           onClick={() => handleTabChange(tab.id)}
261           className={`flex flex-col items-center justify-center min-w-[56px] min-...
262             isActive
263               ? "text-[#00288e] font-bold"
264               : "text-[#444653] hover:text-[#0b1c30]"
265             `}
266         >
267           {/ * Icon with pill bg */}
268           <div
269             className={`px-3 py-1 rounded-full flex items-center justify-center ...
270               isActive
271                 ? "bg-[#dde1ff] scale-110 shadow-xs text-[#00288e]"
272                 : "text-[#444653]"
273             `}
274           >
275             {tab.icon}
276           </div>
277           <span
278             className={`text-[11px] mt-0.5 leading-none transition-all duration-...
279               isActive ? "font-bold text-[#00288e] scale-105" : "text-[#444653]"
280             `}
281           >
282             {tab.label}
283           </span>
284
285           {/ * Animated active indicator dot */}
286           {isActive && (
```

```
287         <span className="w-1.5 h-1.5 rounded-full bg-[#00288e] mt-0.5 animat...
288     })
289 </button>
290 );
291 }}}
292 </div>
293 </nav>
294 </div>
295 );
296 }
```

7.2 Konfigurasi Web App Manifest (public/manifest.json)

```
KODE LENGKAP: public/manifest.json
1  {
2    "name": "E-Presensi Guru",
3    "short_name": "E-Presensi",
4    "description": "Aplikasi Presensi Digital & Absensi Kehadiran Guru Terintegrasi Berb...",
5    "start_url": "/guru",
6    "scope": "/",
7    "display": "standalone",
8    "background_color": "#f8f9ff",
9    "theme_color": "#00288e",
10   "orientation": "portrait",
11   "categories": ["education", "productivity"],
12   "icons": [
13     {
14       "src": "/icons/icon-192x192.png",
15       "sizes": "192x192",
16       "type": "image/png",
17       "purpose": "any maskable"
18     },
19     {
20       "src": "/icons/icon-512x512.png",
21       "sizes": "512x512",
22       "type": "image/png",
23       "purpose": "any maskable"
24     },
25     {
26       "src": "/icons/app-logo.png",
27       "sizes": "512x512",
28       "type": "image/png",
29       "purpose": "any"
30     }
31   ],
32   "shortcuts": [
33     {
34       "name": "Presensi Guru",
35       "short_name": "Presensi",
36       "description": "Buka modul presensi kehadiran guru",
37       "url": "/guru",
38       "icons": [{ "src": "/icons/icon-192x192.png", "sizes": "192x192" }]
39     },
40     {
41       "name": "Login Guru",
42       "short_name": "Login",
43       "description": "Halaman masuk akun GTK",
44       "url": "/guru/login",
45       "icons": [{ "src": "/icons/icon-192x192.png", "sizes": "192x192" }]
46     }
47   ]
48 }
```

7.3 Service Worker Cache Offline (public/sw.js)

```
KODE LENGKAP: public/sw.js
1  // E-Presensi Guru PWA Service Worker
2  const CACHE_NAME = "epresensi-guru-v2";
3  const STATIC_ASSETS = [
4    "/guru/login",
5    "/manifest.json",
6    "/icons/icon-192x192.png",
```

```

7    "/icons/icon-512x512.png",
8    "/icons/app-logo.png",
9    "/icons/apple-touch-icon.png"
10 ];
11
12 // Install event
13 self.addEventListener("install", (event) => {
14     event.waitUntil(
15         caches.open(CACHE_NAME).then((cache) => {
16             return cache.addAll(STATIC_ASSETS);
17         }).catch(() => {})
18     );
19     // Only auto-skipWaiting on initial installation so update prompt can be presented t...
20     if (!self.registration.active) {
21         self.skipWaiting();
22     }
23 });
24
25 // Listen for client message to skip waiting when user confirms update
26 self.addEventListener("message", (event) => {
27     if (event.data && event.data.type === "SKIP_WAITING") {
28         self.skipWaiting();
29     }
30 });
31
32 // Activate event - clean old caches
33 self.addEventListener("activate", (event) => {
34     event.waitUntil(
35         caches.keys().then((keys) => {
36             return Promise.all(
37                 keys.filter((key) => key !== CACHE_NAME).map((key) => caches.delete(key))
38             );
39         })
40     );
41     self.clients.claim();
42 });
43
44 // Fetch event - Network first strategy for dynamic app, with offline cache fallback
45 self.addEventListener("fetch", (event) => {
46     // Only handle standard HTTP and HTTPS requests
47     // Ignore chrome-extension://, moz-extension://, data:, blob:, ws:, etc.
48     if (!event.request.url.startsWith("http://") && !event.request.url.startsWith("https...
49         return;
50     }
51
52     // Only handle GET requests
53     if (event.request.method !== "GET") {
54         return;
55     }
56
57     let url;
58     try {
59         url = new URL(event.request.url);
60     } catch {
61         return;
62     }
63
64     // Bypass API requests, authentication, and Next.js hot reload / webpack
65     if (
66         url.pathname.startsWith("/api/") ||
67         url.pathname.includes("/_next/webpack-hmr")
68     ) {
69         return;
70     }
71
72     // Handle static and page requests
73     event.respondWith(
74         fetch(event.request)
75             .then((response) => {
76                 // Cache successful GET responses for static assets
77                 if (
78                     response &&
79                     response.status === 200 &&
80                     (url.pathname.startsWith("/icons/") ||
81                     url.pathname.endsWith(".png") ||
82                     url.pathname.endsWith(".svg") ||
```

```
83         url.pathname.endsWith(".css") ||
84         url.pathname.endsWith(".js"))
85     ) {
86         const responseToCache = response.clone();
87         caches.open(CACHE_NAME).then((cache) => {
88             cache.put(event.request, responseToCache).catch(() => {});
89         }).catch(() => {});
90     }
91     return response;
92 })
93 .catch(async () => {
94     // Fallback to cache if network fails
95     try {
96         const cachedResponse = await caches.match(event.request);
97         if (cachedResponse) {
98             return cachedResponse;
99         }
100         if (event.request.mode === "navigate") {
101             const loginCache = await caches.match("/guru/login");
102             if (loginCache) return loginCache;
103         }
104     } catch {
105         // ignore cache lookup errors
106     }
107     return new Response("Offline - Tidak ada koneksi internet", {
108         status: 503,
109         statusText: "Service Unavailable",
110         headers: new Headers({ "Content-Type": "text/plain" }),
111     });
112 })
113 );
114 });
```

BAB 8

Core Modul 4: Dashboard Monitoring Admin & Ekspor EMIS GTK 4.0

Panel pemantauan eksekutif madrasah dengan kartu metrik real-time, visualisasi grafik mingguan Recharts, pencatatan absensi massal, serta generator laporan spreadsheet resmi.

8.1 Halaman Utama Dashboard Admin (src/app/(web)/admin/page.tsx)

Kode sumber Server Component halaman utama monitoring admin:

```
KODE LENGKAP: src/app/(web)/admin/page.tsx
1 import React from "react";
2 import { AttendanceStatsCards } from "@features/attendance/components/attendance-stat...
3 import { AttendanceCharts } from "@features/attendance/components/attendance-charts";
4 import { AttendanceTable, type AttendanceRowData } from "@features/attendance/compone...
5 import { GeofenceQuickCard } from "@features/madrasah/components/geofence-quick-card"...
6 import { fetchAdminDashboardData } from "@server/actions/attendance.actions";
7
8 export const dynamic = "force-dynamic";
9
10 export default async function AdminDashboardPage() {
11   const result = await fetchAdminDashboardData();
12   const dashboardData = result?.data;
13
14   const madrasah = dashboardData?.madrasah;
15   const settings = madrasah?.settings;
16   const stats = dashboardData?.stats;
17   const todayLogs = (dashboardData?.todayLogs || []) as unknown as AttendanceRowData[]...
18
19   const presentCount = stats?.presentCount ?? 0;
20   const lateCount = stats?.lateCount ?? 0;
21   const permitCount = stats?.permitCount ?? 0;
22   const sickCount = stats?.sickCount ?? 0;
23   const absentCount = stats?.absentCount ?? 0;
24   const totalTeachers = stats?.totalTeachers ?? 0;
25   const weeklyData = (dashboardData as any)?.weeklyData;
26   const geofenceZones = (dashboardData as any)?.geofenceZones;
27
28   return (
29     <div className="flex flex-col gap-6 w-full">
30       {/* 1. KPI Metric Summary Cards */}
31       <AttendanceStatsCards
32         totalTeachers={totalTeachers}
33         presentCount={presentCount}
34         lateCount={lateCount}
35         permitCount={permitCount}
36         sickCount={sickCount}
37         absentCount={absentCount}
38         attendancePercentage={stats?.attendancePercentage ?? 0}
39         radiusMeters={settings?.radiusMeters || 50}
40         latitude={settings?.latitude || -6.2615}
41         longitude={settings?.longitude || 106.8106}
42       />
43
44       {/* 2. Visual Charts & Informative Widgets (Weekly Trend, Donut Composition, Geo...
45       <AttendanceCharts
46         presentCount={presentCount}
47         lateCount={lateCount}
48         permitCount={permitCount}
49         sickCount={sickCount}
50         totalTeachers={totalTeachers}
51         radiusMeters={settings?.radiusMeters || 50}
52         weeklyData={weeklyData}
53         geofenceZones={geofenceZones}
54         workStartTime={settings?.workStartTime}
55         lateThreshold={settings?.lateThreshold}
56         workEndTime={settings?.workEndTime}
57       />
58     )
59 }
```

```
59      {/* 3. Geofence Parameters Card */}
60      <GeofenceQuickCard
61        latitude={settings?.latitude || -6.2615}
62        longitude={settings?.longitude || 106.8106}
63        radiusMeters={settings?.radiusMeters || 50}
64        workStartTime={settings?.workStartTime || "07:00"}
65        lateThreshold={settings?.lateThreshold || "07:15"}
66      />
67
68      {/* 4. Interactive Attendance Datatable */}
69      <AttendanceTable
70        data={todayLogs}
71        maxRadiusMeters={settings?.radiusMeters || 50}
72      />
73    </div>
74  );
75 }
```

8.2 Repository Data Presensi Admin (src/server/repositories/attendance.repo.ts)

Fungsi database teroptimasi untuk menghitung metrik KPI dan daftar hadir harian:

```
KODE LENGKAP: src/server/repositories/attendance.repo.ts (Salinkan Utama)
1  import { prisma } from "@lib/prisma";
2
3  export interface AttendanceFilters {
4    date?: Date;
5    status?: string;
6    search?: string;
7  }
8
9  export async function getMadrasahDashboardData(madrasahId: string) {
10    const now = new Date();
11    const today = new Date(Date.UTC(now.getFullYear(), now.getMonth(), now.getDate(), 0, ...
12    const tomorrow = new Date(Date.UTC(now.getFullYear(), now.getMonth(), now.getDate() ...
13
14    // 1. Get Madrasah & Geofence settings
15    const madrasah = await prisma.madrasah.findUnique({
16      where: { id: madrasahId },
17      include: {
18        settings: true,
19      },
20    });
21
22    // 2. Total active teachers
23    const totalTeachers = await prisma.user.count({
24      where: {
25        madrasahId,
26        role: "TEACHER",
27        isActive: true,
28      },
29    });
30
31    // 3. Today's attendance logs (optimized select to prevent massive base64 photo over...
32    const todayLogs = await prisma.attendanceLog.findMany({
33      where: {
34        madrasahId,
35        date: {
36          gte: today,
37          lt: tomorrow,
38        },
39      },
40      select: {
41        id: true,
42        userId: true,
43        date: true,
44        status: true,
45        checkInTime: true,
46        checkInDistance: true,
47        checkInLat: true,
48        checkInLng: true,
49        checkInPhotoUrl: true,
50        notes: true,
51        user: {
```

```
52     select: {
53       id: true,
54       name: true,
55       nip: true,
56       email: true,
57       avatarUrl: true,
58     },
59   },
60 },
61   orderBy: {
62     checkInTime: "asc",
63   },
64 });
65
66 // 4. Calculate KPI statistics
67 const presentCount = todayLogs.filter((l) => l.status === "PRESENT").length;
68 const lateCount = todayLogs.filter((l) => l.status === "LATE").length;
69 const permitCount = todayLogs.filter((l) => l.status === "PERMIT").length;
70 const sickCount = todayLogs.filter((l) => l.status === "SICK").length;
71 const absentCount = Math.max(
72   0,
73   totalTeachers - (presentCount + lateCount + permitCount + sickCount)
74 );
75
76 // 5. Weekly trend data for Monday - Friday of the current week from database
77 const dayOfWeek = now.getDay(); // 0 = Sun, 1 = Mon, ..., 6 = Sat
78 const mondayOffset = dayOfWeek === 0 ? -6 : 1 - dayOfWeek;
79 const monday = new Date(now);
80 monday.setDate(now.getDate() + mondayOffset);
81
82 const startOfWeek = new Date(Date.UTC(monday.getFullYear(), monday.getMonth(), monda...
83 const endOfWeek = new Date(Date.UTC(monday.getFullYear(), monday.getMonth(), monday...
84
85 const weeklyLogs = await prisma.attendanceLog.findMany({
86   where: {
87     madrasahId,
88     date: {
89       gte: startOfWeek,
90       lte: endOfWeek,
91     },
92   },
93   select: {
94     id: true,
95     date: true,
96     status: true,
97   },
98 });
99
100 const dayNames = ["Senin", "Selasa", "Rabu", "Kamis", "Jumat"];
101 const weeklyData = dayNames.map((name, idx) => {
102   const curDate = new Date(monday);
103   curDate.setDate(monday.getDate() + idx);
104   const curDateStr = `${curDate.getFullYear()}-${String(curDate.getMonth() + 1).padS...
105
106   const logsForDay = weeklyLogs.filter((l) => {
107     const logDateStr = new Date(l.date).toISOString().split("T")[0];
108     return logDateStr === curDateStr;
109   });
110
111   const hadir = logsForDay.filter((l) => l.status === "PRESENT").length;
112   const terlambat = logsForDay.filter((l) => l.status === "LATE").length;
113   const izin = logsForDay.filter((l) => l.status === "PERMIT" || l.status === "SICK"...
114
115   return {
116     day: name,
117     hadir,
118     terlambat,
119     izin,
120   };
121 });
122
123 // 6. Real Geofence breakdown from today's attendance logs
124 const maxRadius = madrasah?.settings?.radiusMeters || 50;
125 let zoneInti = 0;
126 let zoneLuar = 0;
127 let zoneKritis = 0;
```

```
128     let zoneLuarRadius = 0;
129
130     todayLogs.forEach((l) => {
131         const dist = l.checkInDistance;
132         if (dist !== null && dist !== undefined) {
133             if (dist <= 25) {
134                 zoneInti++;
135             } else if (dist <= 40) {
136                 zoneLuar++;
137             } else if (dist <= maxRadius) {
138                 zoneKritis++;
139             } else {
140                 zoneLuarRadius++;
141             }
142         }
143     });
144
145     const totalLogsCount = todayLogs.length || 1;
146     const geofenceZones = [
147         {
148             label: "Zona Inti (0 - 25m)",
149             count: `${zoneInti} Guru`,
150             percent: todayLogs.length > 0 ? Math.round((zoneInti / totalLogsCount) * 100) : ...
151             color: "bg-emerald-500",
152         },
153         {
154             label: "Zona Luar (25 - 40m)",
155             count: `${zoneLuar} Guru`,
156             percent: todayLogs.length > 0 ? Math.round((zoneLuar / totalLogsCount) * 100) : ...
157             color: "bg-teal-500",
158         },
159         {
160             label: `Batas Kritis (40 - ${maxRadius}m)`,
161             count: `${zoneKritis} Guru`,
162             percent: todayLogs.length > 0 ? Math.round((zoneKritis / totalLogsCount) * 100) ...
163             color: "bg-amber-500",
164         },
165         {
166             label: `Di Luar Radius (> ${maxRadius}m)`,
167             count: `${zoneLuarRadius} Guru`,
168             percent: todayLogs.length > 0 ? Math.round((zoneLuarRadius / totalLogsCount) * 1...
169             color: "bg-rose-500",
170         },
171     ];
172
173     return {
174         madrasah,
175         stats: {
176             totalTeachers,
177             presentCount,
178             lateCount,
179             permitCount,
180             sickCount,
181             absentCount,
182             attendancePercentage:
183                 totalTeachers > 0
184                 ? Math.round(((presentCount + lateCount) / totalTeachers) * 100)
185                 : 0,
186         },
187         todayLogs,
188         weeklyData,
189         geofenceZones,
190     };
191 }
192
193 export interface TeacherWithAttendance {
194     id: string;
195     name: string;
196     email: string;
197     nip: string | null;
198     phone: string | null;
199     avatarUrl: string | null;
200     isActive: boolean;
201     attendanceLog: {
202         id: string;
203         status: "PRESENT" | "LATE" | "PERMIT" | "SICK" | "ABSENT";
```

204	checkInTime: Date null;
205	checkInDistance: number null;
206	checkInLat: number null;
207	checkInLng: number null;
208	checkOutTime: Date null;
209	notes: string null;
210	date: Date;
211	} null;
212	}
213	
214	export interface DateHolidayInfo {
215	isHoliday: boolean;
216	type: "SUNDAY" "HOLIDAY" null;
217	name:

BAB 9

PANDUAN PRAKTIK: Menambahkan Modul Baru dari Nol (Studi Kasus)

Panduan praktis langkah demi langkah (step-by-step masterclass) membuat fitur/modul baru dari skema database, logika backend, antarmuka front-end, hingga pendaftaran menu di navigasi.

STUDI KASUS NYATA: MODUL SURAT TUGAS & SPPD GURU

Kita akan mempraktikkan secara tuntas pembuatan modul baru: 'Manajemen Surat Tugas GTK (SPPD)'. Modul ini berfungsi mencatat penugasan luar dinas guru (seminar, pelatihan Kemenag, workshop) yang terhubung ke data master guru dan madrasah.

Langkah 1: Tambahkan Model Data di prisma/schema.prisma

Buka `prisma/schema.prisma` dan tambahkan model relasional `DutyLetter` berikut:

```
KODE LENGKAP: prisma/schema.prisma (Model Baru)
1  model DutyLetter {
2    id          String      @id @default(uuid())
3    madrasahId  String      @map("madrasah_id")
4    userId      String      @map("user_id")
5    letterNumber String      @map("letter_number") // Contoh: B-102/Mi.01/PP.00/10/2026
6    purpose     String      @db.Text                // Maksud & Tujuan Tugas
7    destination String      @db.Text                // Tempat / Lokasi Penugasan
8    startDate   DateTime    @db.Date @map("start_date")
9    endDate     DateTime    @db.Date @map("end_date")
10   status      String      @default("ACTIVE")      // ACTIVE / COMPLETED / CANCELLED
11   attachmentUrl String?   @map("attachment_url")
12   createdAt   DateTime    @default(now()) @map("created_at")
13   updatedAt   DateTime    @updatedAt @map("updated_at")
14
15   madrasah     Madrasah   @relation(fields: [madrasahId], references: [id], onDelete: ...
16   user         User       @relation(fields: [userId], references: [id], onDelete: Casc...
17
18   @@index([madrasahId])
19   @@index([userId])
20   @@index([madrasahId, startDate, endDate])
21   @@map("duty_letters")
22 }
```

Langkah 2: Sinkronisasi Skema Database

```
KODE LENGKAP: terminal / powershell
1  # Jalankan migrasi dan generate Prisma Client
2  npx prisma db push
3  npx prisma generate
```

Langkah 3: Buat Server Action (src/server/actions/duty-letter.actions.ts)

```
KODE LENGKAP: src/server/actions/duty-letter.actions.ts
1  "use server";
2
3  import { prisma } from "@/lib/prisma";
4  import { revalidatePath } from "next/cache";
5  import { requireMadrasahAdmin } from "@/server/utils/auth-guard";
6
7  export interface DutyLetterPayload {
8    userId: string;
9    letterNumber: string;
10   purpose: string;
11   destination: string;
12   startDate: string;
13   endDate: string;
```

```
14 }
15
16 export async function getDutyLettersAction() {
17   try {
18     const { madrasahId } = await requireMadrasahAdmin();
19     const data = await prisma.dutyLetter.findMany({
20       where: { madrasahId },
21       include: {
22         user: {
23           select: { id: true, name: true, nip: true, avatarUrl: true },
24         },
25       },
26       orderBy: { startDate: "desc" },
27     });
28     return { success: true, data };
29   } catch (error: any) {
30     return { error: error.message || "Gagal memuat surat tugas." };
31   }
32 }
33
34 export async function createDutyLetterAction(payload: DutyLetterPayload) {
35   try {
36     const { madrasahId } = await requireMadrasahAdmin();
37     if (!payload.userId || !payload.letterNumber || !payload.purpose) {
38       return { error: "Semua bidang formulir wajib diisi." };
39     }
40
41     const created = await prisma.dutyLetter.create({
42       data: {
43         madrasahId,
44         userId: payload.userId,
45         letterNumber: payload.letterNumber.trim(),
46         purpose: payload.purpose.trim(),
47         destination: payload.destination.trim(),
48         startDate: new Date(payload.startDate),
49         endDate: new Date(payload.endDate),
50       },
51     });
52
53     revalidatePath("/admin/duty-letters");
54     return { success: true, data: created };
55   } catch (error: any) {
56     return { error: error.message || "Gagal menyimpan surat tugas baru." };
57   }
58 }
59
60 export async function deleteDutyLetterAction(id: string) {
61   try {
62     const { madrasahId } = await requireMadrasahAdmin();
63     await prisma.dutyLetter.deleteMany({
64       where: { id, madrasahId }, // Isolasi multi-tenant terjamin
65     });
66     revalidatePath("/admin/duty-letters");
67     return { success: true };
68   } catch (error: any) {
69     return { error: error.message || "Gagal menghapus surat tugas." };
70   }
71 }
```

Langkah 4: Buat Komponen Tampilan Klien (src/features/duty-letters/components/duty-letters

```
KODE LENGKAP: src/features/duty-letters/components/duty-letters-view.tsx
1  "use client";
2
3  import React, { useState } from "react";
4  import { Plus, Trash2, FileText, Calendar, MapPin, User } from "lucide-react";
5  import { createDutyLetterAction, deleteDutyLetterAction } from "@/server/actions/duty-...
6  import { swalSuccess, swalError } from "@/lib/swal";
7
8  export function DutyLettersView({ initialData, teachers }: { initialData: any[]; teach...
9    const [letters, setLetters] = useState(initialData);
10    const [isOpenModal, setIsOpenModal] = useState(false);
11    const [form, setForm] = useState({
12      userId: teachers[0]?.id || "",
```

```
13     letterNumber: "",
14     purpose: "",
15     destination: "",
16     startDate: new Date().toISOString().split("T")[0],
17     endDate: new Date().toISOString().split("T")[0],
18   });
19
20   const handleSubmit = async (e: React.FormEvent) => {
21     e.preventDefault();
22     const res = await createDutyLetterAction(form);
23     if (res.error) {
24       swalError("Gagal", res.error);
25       return;
26     }
27     swalSuccess("Berhasil", "Surat tugas berhasil diterbitkan!");
28     setIsOpenModal(false);
29     window.location.reload();
30   };
31
32   return (
33     <div className="flex flex-col gap-6 w-full">
34       <div className="flex items-center justify-between">
35         <div>
36           <h2 className="text-xl font-bold text-slate-900">Surat Tugas & Perjalanan Di...
37           <p className="text-xs text-slate-500">Daftar penugasan luar dinas resmi guru...
38         </div>
39         <button onClick={() => setIsOpenModal(true)} className="btn-primary flex items...
40           <Plus className="w-4 h-4" /> Terbitkan Surat Tugas
41         </button>
42       </div>
43       { /* Tabel Data & Modal Tambah Data */ }
44     </div>
45   );
46 }
```

Langkah 5: Buat Halaman Route Next.js (src/app/(web)/admin/duty-letters/page.tsx)

```
KODE LENGKAP: src/app/(web)/admin/duty-letters/page.tsx
1  import React from "react";
2  import { getDutyLettersAction } from "@server/actions/duty-letter.actions";
3  import { fetchTeachersData } from "@server/actions/teacher.actions";
4  import { DutyLettersView } from "@features/duty-letters/components/duty-letters-view"...
5
6  export const dynamic = "force-dynamic";
7  export const metadata = { title: "Surat Tugas GTK | E-Presensi Admin" };
8
9  export default async function DutyLettersPage() {
10    const lettersRes = await getDutyLettersAction();
11    const teachersRes = await fetchTeachersData();
12
13    return (
14      <DutyLettersView
15        initialData={lettersRes.data || []}
16        teachers={teachersRes.data || []}
17      />
18    );
19  }
```

Langkah 6: Daftarkan Menu pada Sidebar Navigasi Admin

Buka `src/components/organisms/sidebar.tsx` dan tambahkan menu baru pada kelompok menu yang sesuai:

```
KODE LENGKAP: src/components/organisms/sidebar.tsx (Penambahan Item Menu)
1  {
2    label: "Surat Tugas (SPPD)",
3    href: "/admin/duty-letters",
4    icon: FileSpreadsheet, // Import ikon dari lucide-react
5    badge: "Baru"
6  }
```

BAB 10

Audit Performa, Optimasi Skema, Keamanan & Deployment Produksi

Daftar periksa optimasi latensi database, eliminasi overfetching foto Base64, konfigurasi next.config.ts, audit keamanan data, serta panduan deployment ke VPS Ubuntu atau Vercel.

10.1 Konfigurasi next.config.ts Teroptimasi

Konfigurasi optimasi package imports untuk mempercepat bundle dan First Load JS:

```
KODE LENGKAP: next.config.ts
1 import type { NextConfig } from "next";
2
3 const nextConfig: NextConfig = {
4   experimental: {
5     optimizePackageImports: ["lucide-react", "recharts"],
6   },
7 };
8
9 export default nextConfig;
```

10.2 Konfigurasi Web Server Nginx (Reverse Proxy & SSL)

```
KODE LENGKAP: /etc/nginx/sites-available/epresensi
1 server {
2     server_name presensi.madrasah.sch.id;
3
4     location / {
5         proxy_pass http://127.0.0.1:3000;
6         proxy_http_version 1.1;
7         proxy_set_header Upgrade $http_upgrade;
8         proxy_set_header Connection 'upgrade';
9         proxy_set_header Host $host;
10        proxy_cache_bypass $http_upgrade;
11        proxy_set_header X-Real-IP $remote_addr;
12        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
13        proxy_set_header X-Forwarded-Proto $scheme;
14    }
15
16    listen 443 ssl http2;
17    ssl_certificate /etc/letsencrypt/live/presensi.madrasah.sch.id/fullchain.pem;
18    ssl_certificate_key /etc/letsencrypt/live/presensi.madrasah.sch.id/privkey.pem;
19 }
```

10.3 Konfigurasi Process Manager PM2 (ecosystem.config.js)

```
KODE LENGKAP: ecosystem.config.js
1 module.exports = {
2   apps: [
3     {
4       name: 'epresensi-production',
5       script: 'node_modules/next/dist/bin/next',
6       args: 'start',
7       cwd: './web',
8       instances: 'max',
9       exec_mode: 'cluster',
10      max_memory_restart: '1G',
11      env: {
12        NODE_ENV: 'production',
13        PORT: 3000,
14      },
15    },
16  ],
17};
```

10.4 Skrip Otomatisasi Backup Database PostgreSQL (backup-db.sh)

```
KODE LENGKAP: eprints/backup-db.sh
1  #!/bin/bash
2  TIMESTAMP=$(date +%Y%m%d_%H%M%S")
3  BACKUP_DIR="/var/backups/epresensi"
4  mkdir -p $BACKUP_DIR
5
6  # Dump database PostgreSQL terkompresi
7  pg_dump -U postgres -h aws-0-ap-northeast-1.pooler.supabase.com -p 5432 postgres | gzi...
8
9  # Hapus file backup yang berusia lebih dari 30 hari
10 find $BACKUP_DIR -type f -name "*.sql.gz" -mtime +30 -exec rm {} \;
11 echo "[OK] Backup database selesai pada $TIMESTAMP"
```